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	Particle	
	Given Name	Atef
	Suffix	
	Division	Department of Agricultural and Biosystems Engineering
	Organization	Alexandria University
	Address	Alexandria, Egypt
Corresponding Author	Family Name	Bekhit
	Particle	
	Given Name	Alaa El-Din A.
	Suffix	
	Division	Department of Food Science
	Organization	University of Otago
	Address	PO Box 56, Dunedin, 9054, New Zealand
	Email	aladin.bekhit@otago.ac.nz
Abstract	Over the history of mankind, food preservation has played a crucial role in alleviating hunger and famine when food was less abundant and was instrumental in supplying edible foods for armies, voyagers and explorations that led to significant changes in human history. Progress in food preservation and processing can be considered as four main processing paradigm shifts over human history: (1) production of safe foods (early human trials–1950s); (2) production of safe food that have desirable sensory properties, i.e. looks good and tastes good, (1960s–1980s); (3) production of safe high quality food that is nutritionally sound (1990s–2000s); and 4) production of safe nutritious high quality food that delivers specific functions (2000s–present). Hence, the food industry moved from primitive concepts, such as simple dehydrated products and cured foods, to technologies such as canning, refrigeration and freezing. The purpose was to meet the demands for healthy and nutritious safe foods.	

Chapter 7

Processing, Storage and Quality of Cook-Chill or Cook-Freeze Foods

Atef El-Ansari and Alaa El-Din A. Bekhit

7.1 Introduction

Over the history of mankind, food preservation has played a crucial role in alleviating hunger and famine when food was less abundant and was instrumental in supplying edible foods for armies, voyagers and explorations that led to significant changes in human history. Progress in food preservation and processing can be considered as four main processing paradigm shifts over human history: (1) production of safe foods (early human trials–1950s); (2) production of safe food that have desirable sensory properties, i.e. looks good and tastes good, (1960s–1980s); (3) production of safe high quality food that is nutritionally sound (1990s–2000s); and 4) production of safe nutritious high quality food that delivers specific functions (2000s–present). Hence, the food industry moved from primitive concepts, such as simple dehydrated products and cured foods, to technologies such as canning, refrigeration and freezing. The purpose was to meet the demands for healthy and nutritious safe foods.

The increased demand for ready to eat fast meals prepared by food catering sectors outside the home during the 1960s highlighted the value of cooked-chilled and cooked-frozen foods. Significant interest in the 1960s and 1970s resulted in implementation of the technology at the commercial or industrial scale. The interest in technology development decreased during the 1970s and the 1980s due to high production costs associated with cost of electricity, labour and the capital cost of equipment. Total food expenditure on the meals prepared outside the home was

A. El-Ansari
Department of Agricultural and Biosystems Engineering,
Alexandria University, Alexandria, Egypt

A.E.-D.A. Bekhit (✉)
Department of Food Science, University of Otago, PO Box 56,
Dunedin 9054, New Zealand
e-mail: aladin.bekhit@otago.ac.nz

reported to increase about twofold in the US during the 1990s (Manchester and Clauson 1994). This trend of expenditure as well as other demographic changes in working hours and increased participation of women in the labour force (Jekanowski et al. 2001), increased the value of convenience (Hiemstra and Kim 1995) and promoted the availability of more energy-efficient processing systems (Light and Walker 1990), which revived interest in cook-chill and cook-frozen technologies in the food industry.

7.2 Cook-Chill and Cook-Freeze Systems

A cook-chill system is a preservation technology where food (raw material and other ingredients) is fully cooked at usually $<100\text{ }^{\circ}\text{C}$ and the product is subsequently chilled rapidly and stored under controlled conditions where the final core temperature is maintained above the freezing point at $0\text{--}3\text{ }^{\circ}\text{C}$ and can produce an extended shelf life at $4\text{--}10\text{ }^{\circ}\text{C}$ (Evans et al. 1996; Sun and Hu 2002). The use of high quality raw material is an essential requirement to guarantee high quality products. The destruction and inhibition of microorganisms, especially the pathogenic ones, are achieved through the initial cooking step and subsequent storage at lower temperatures (Hill 1994). The general cook-chill process in Denmark, France, Germany, Sweden and the UK requires a thermal treatment in the range of $65\text{--}80\text{ }^{\circ}\text{C}$ followed by a chilling process to achieve a temperature of $10\text{--}3\text{ }^{\circ}\text{C}$ within 1.5–4 h and then stored at a temperature within $0\text{--}3\text{ }^{\circ}\text{C}$ (Hill 1994). More recently “supercooling conditions” are being used to extend the shelf-life of the products by maintaining their temperature at $1\text{--}2\text{ }^{\circ}\text{C}$ below freezing temperature.

The cook-chill process is also conducted under vacuum (referred to as ‘sous-vide’; Fig. 7.1). In this process, food is vacuum packed inside a heat-stable pouch or tray before cooking (Nyati 2000). Sous-vide products are normally cooked at a temperature of less than $80\text{ }^{\circ}\text{C}$ and a shelf-life of up to 28 days is usually achieved (Tansey et al. 2010). The extended shelf life is limited to 28 days due to the potential of toxin production by non-proteolytic strains of *Clostridium botulinum* at $3.3\text{ }^{\circ}\text{C}$ within 31 days (Schmidt et al. 1961). The vacuum bag barrier between the cooked product and the environment protects the product from potential post-cooking contamination.

Cook-freeze food products are treated the same way as cook-chill products, but the final product is either super-chilled (i.e. cooled and maintained at $1\text{--}2\text{ }^{\circ}\text{C}$ below the freezing temperature of the product) or frozen-chilled (i.e. frozen at a temperature of below $-30\text{ }^{\circ}\text{C}$ and then thawed to $5\text{ }^{\circ}\text{C}$ when needed; O’Leary et al. 2000; Redmond et al. 2004). This allows ‘chilled’ foods’ delivery to more distant markets, facilitates bulk production, and reduces the level of product recalls. The main limitation of the shelf life of cook-chill and sous-vide processed products are the degradation of sensory attributes and microbial growth. These attributes depend on



Fig. 7.1 Examples of some commercial cook-chill and sous-vide products

the storage conditions (time and temperature). Diaz et al. (2008) investigated the spoilage over 10 weeks of sous-vide pork loin at chilled condition (i.e. cooked at 70 °C for 12 h, and then chilled at 3 °C) by examining microbial, physicochemical and sensory attributes. Their results demonstrated that sensory spoilage occurred prior to microbial spoilage and the pork was unacceptable after 10 weeks of storage based on its sensory properties. Thus this method provides extended protection against microbial concerns.

The shelf-life of cook-freeze products is limited by structural changes in the food (i.e. ice crystal size and off flavour formation). This is due to the production of chemicals, such as formaldehyde and free fatty acids by hydrolysis and oxidation. The shelf life of cook-freeze products is dependent on the chemical composition of the food, packaging system and storage conditions (i.e. time and temperature). Foods processed using the above technologies are collectively known as refrigerated processed foods (REFPED; Rajkovic et al. 2010).

Several recommendations have been adopted in regulations and guidelines for the production of cook-chill and sous-vide food products include: US Food Code 2003 in 3-502.11.D and 3-502.12, (with additional information in Annex 3, 4 and 6) <http://www.fda.gov/downloads/Food/GuidanceRegulation/RetailFoodProtection/FoodCode/UCM374510.pdf>; European Chilled Food Federation recommendations http://www.ecff.net/images/ECFF_Recommendations_2nd_ed_18_12_06.pdf; Food safety authority of Ireland recommendations www.fsai.ie/WorkArea/DownloadAsset.aspx?id=746; NSW Food authority Australia http://www.foodauthority.nsw.gov.au/_Documents/industry_pdf/guidelines_vp_2011.pdf, UK <http://www.food.gov.uk/multimedia/pdfs/csctcooking.pdf>.

7.3 Microbiological Considerations

The processing associated with REPFED (Table 7.1) is based on the requirement of a 6 log reduction of *L. monocytogenes* and *C. botulinum*. This is regarded as a “worst case scenario” in chilled products (Rybka 2000; Rybka et al. 2001). The products are processed at a temperature range of 65–100 °C for different times. Maintaining the center of a product at 60 °C for 45 min, 65 °C for 10 min or at 70 °C for 2 min can deliver the same lethality value (Rajkovic et al. 2010). Even lower temperatures with longer treatment times may be acceptable (the ‘thermal death concept’). The use of lower temperatures may be preferred to maintain the sensory properties of a product or to enable faster cooling, but this can lead to post-processing problems. The raw material and personal hygiene can be a source of unwanted aerobic microorganisms (e.g. *Staphylococcus aureus* and *Bacillus cereus*, respectively) and this can represent a significant health hazard. Thus the importance of good hygiene and a clean production system needs to be emphasized. The use of mild cooking treatments (e.g. pasteurization) in cook-chill products may not be sufficient to inactivate pathogenic spores and therefore a low chilling rate during post-cooking may pose a potential problem with spores germinating to vegetative cells. Bacteria can potentially survive mild heat treatment, for examples non-proteolytic *Clostridium botulinum* (Anaerobic, $D_{80^{\circ}\text{C}} = 21.6$ min, $z = 7$ °C,

Table 7.1 Processing requirements and shelf life of cook-chill, sous-vide and cook-freeze products

Process step	Cook-chill	Sous-vide	Cook-freeze
Cooking	≥ 70 °C <100 °C for >2 min ^a	≥ 70 °C <100 °C. no time specified	≥ 70 °C <100 °C for >2 min
Time to cooling	Within 30 min	Immediately	Within 30 min
Cooling rate	To 3 °C within 1.5 h	To 3 °C within 1.5 h	To –5 °C at the centre within 1.5 h
Storage temperature	0–3 °C	0–3 °C or –18 °C	≤ -18 °C
Shelf life	5 days at 0–3 °C	8 weeks at –18 °C	8 weeks at –18 °C
	24 h at 5–10 °C	4 weeks at 0–3 °C	
		12 h at 5–10 °C	
Consumptions	Re-heat at 70 °C minimum for not less than 2 min. Consume immediately	Re-heat at 70 °C minimum for not less than 2 min. Consume immediately	Once thawed should be consumed after heating at 70 °C minimum for not less than 2
	Meat joints should not exceed 3 kg or 100 mm thick		

Modified from James and James (2005)

^aThermal treatment (70 °C for 2 min or equivalent) is sufficient to ensure a 6 log reduction in numbers of *L. monocytogenes* which is considered the most heat resistant vegetative pathogen (Gaze et al. 1989)

grows at 3 °C), proteolytic *C. botulinum* (Anaerobic, $D_{121.1^{\circ}\text{C}}=0.23\text{--}0.3$ min, $z=10^{\circ}\text{C}$, grows at $>10^{\circ}\text{C}$), *C. perfringens* (Anaerobic, $D_{98^{\circ}\text{C}}=21\text{--}31$ min, $z=3.8^{\circ}\text{C}$, grows at $>15^{\circ}\text{C}$) and *B. cereus* (Aerobic, $D_{100^{\circ}\text{C}}=2.7\text{--}3.1$ min, $z=6.1^{\circ}\text{C}$, grows at $>4^{\circ}\text{C}$) (Baldwin 2012). *Bacillus cereus* is capable of growing at low temperatures but the growth of the bacteria is significantly lowered at 7 °C. These bacteria are considered to be potentially hazardous in the case of inadequately prepared sous-vide products (Rybka et al. 2001), but strict chilling temperature control below 4 °C is reported to control the growth of *Bacillus* spp. (Carlin et al. 1999). Rapid chilling can control the hazard posed by *C. perfringens* and controlled chilled storage and reheating at $>70^{\circ}\text{C}$ can control *C. botulinum* (Wilkinson et al. 1991). Some incidences of high total aerobic viable count in cook-chill meals prepared in a hospital (Chudasama et al. 1991) were probably related to inadequate re-heating and cross contamination. The potential problem arises if hazard analysis critical control point (HACCP) plans are not in place. The incorporation of additional safety hurdles [irradiation, modified atmosphere packaging, high pressure, use of chemicals (e.g. lactate, pH and salts) and biological agents (e.g. bacteriocins and protective/probiotic cultures)] can potentially be useful to improve the safety of products. However this may cause side effects on the sensory and nutritional properties of the products (Rybka-Rodgers 2001). Bacteriocins and protective cultures can be more acceptable by consumers compared to other technologies, such as irradiation. This is due to their natural image, wide use in food products such as dairy products, and their role in gut health (Rodgers 2004). The use of protective cultures show many beneficial effects (i.e. potential of decreasing the severity of processing; improving the negative effects of heat on thermo-sensitive compounds/foods, or improving the sensory properties of the food; Rodgers 2003). New developments in microencapsulation or production of cultures with high thermal tolerance may be key for significant use of protective cultures in REPFED. Lactic acid bacteria can survive under the storage conditions of sous-vide and cause spoilage (Carlin et al. 1999). Lactic acid bacteria were isolated from a number of sous-vide products, such as spaghetti and meat sauce, chicken breast meat, chicken mousse and chicken chasseur, fish banding and meatballs (Simpson et al. 1994; Nyati 2000). The low pH of the products was thought to aid bacterial growth, but it was not the case with low pH fish curry (Shakila et al. 2012) or sous-vide fish cake (Shakila et al. 2009).

7.4 Applications of Cook-Chill Processing Systems

Cook-chill systems are used mainly in foods from the catering service industry as well as chilled ready meals from supermarkets. Chill-cook foods are widely available in commercial catering businesses such as hotels, restaurants, and fast food outlets with application in the preparation of large numbers of meals needed to be served in a short time, such as universities, airlines, and the army and for “heat and eat” meals. The technology has proven to be successful since there are few reports of food poisoning or food outbreaks due to the consumption of sous-vide, cook-chill or frozen-chill foods (Peck et al. 2006).

7.5 Advantages and Disadvantages of Cook-Chill and Cook-Freeze Products

Cook-chill technology shows great synergy when it is used with irradiation to inhibit *L. monocytogenes* and *Salmonella typhimurium* (Grant and Patterson 1995). This is due to irradiation-induced damage to the bacteria making them more vulnerable to thermal inactivation. Irradiation of cook-chill roast beef at 0.8 kGy can decrease the thermal *D* and *z* values of bacteria by >50 %. This suggests that *L. monocytogenes* and *S. typhimurium* can be easily killed during the reheating step required before serving cook-chill products. Sous-vide microwaving was found to be one of the best treatments to decrease the microbial counts of psychrophilic and enterobacteria on broccoli (Martínez-Hernández et al. 2013). The integration of the cook-chill system with other quality control systems can produce excellent results. For example, the introduction of a HACCP system in hospitals using a cook-chill system for catering resulted in more than 90 % reduction of microbial load ($<1 \times 10^3$ colony forming units (CFU) per g) and complete elimination of pathogenic bacteria (Shanaghy et al. 1993).

Due to the nature of the food catering industry that requires serving a large number of meals in a timely manner and the convenience of ready to eat chilled meals, cook-chill is a more efficient system to meet these demands compared to the cook-serve operation. Cook-chill and cook-freeze technologies provide several advantages over conventional food processing. These can be summarized as follows:

1. Ability to maintain a pre-stocked inventory that eliminates problems during peak demand and shortage of raw materials' supply.
2. Shelf life extension and ability to supply to distant markets.
3. Ability to reduce labour, raw material and other costs by 20–50 % since the food is prepared in advance.
4. Ability to produce larger quantities, which improves the cost and efficiency of production. Although reports claim up to 50 % reduction in the number of FTEs (Chater 2000), some studies reported little difference among conventional-cook, cook-chill, and cook-freeze systems in terms of operation and costs (Greathouse et al. 1989).
5. Provide better control of food production and ability to utilize technologies that provide desirable results (e.g. irradiation or pulsed electric field) since these may have limitations under a lower production scale.
6. Ability to reduce food waste since by-products/wastes can be used. Unlike conventional cooking (cook-serve system), cook-chill and related systems have the ability to cover any un-forecasted demand or losses.

In addition to the advantages mentioned above, Baldwin (2012) described several other advantages for sous-vide technology as follows:

1. Ability to control temperature precisely. This allows product consistency.
2. Temperature control can be translated into a better control over the product sensory properties (e.g. texture, colour) compared to traditional cooking methods.
3. Possible improvements to the tenderness of tough cuts of meat.



Fig. 7.2 Cook-chill central facility

Despite the numerous advantages that cook-chill, cook-freeze and sous-vide systems have the following disadvantages:

1. Requires specialized expensive equipment (Fig. 7.2).
2. Needs strict cold storage and cold chain requirements.
3. Requires thawing steps for freeze-chill foods.
4. Needs fast cooling without surface freezing.
5. In the cases of large packs, cooking continues inside the pack since cooling is not instantaneous. The use of heat exchangers may be useful to overcome this problem for pumpable materials, but it is not suitable for solid foods (i.e. meat, fish and large chunks of vegetables)
6. It may not be suitable for all types of foods and cuisines. For examples, grills and caramelized products cannot be generated using these technologies.

7.6 Processing of Cook-Chill/Sous-Vide and Cook-Freeze Products

A general layout of the cook-chill system is shown in Fig. 7.3. This demonstrates a series of steps or stages and each individual step is regarded equally important in order to maintain both safety and to produce good quality products. Cook-chill technique is simple, and shows flexibility in food service. The methodology involves the full cooking of food, followed by fast chilling and storage at cool temperatures. This can extend shelf life up to 5 days. Within the recommended storage period, the prepared foods can be served quickly after a re-heating step when stored within the recommended storage time. The production system itself is simple to operate and provides a consistent quality. A safe product can be obtained if recommended temperature/time controls are followed. Normally, the recommended minimum time required is 2 min at core (centre) temperature of 70 °C. The temperature of the product should be rapidly reduced from 60 to 7 °C to prevent any growth of certain micro-organisms. Further reduction to 3 °C is essential to reduce the growth of spoilage bacteria and pathogens. The main difference between cook-chill and cook-freeze is their final core temperature and the required cooling rates.

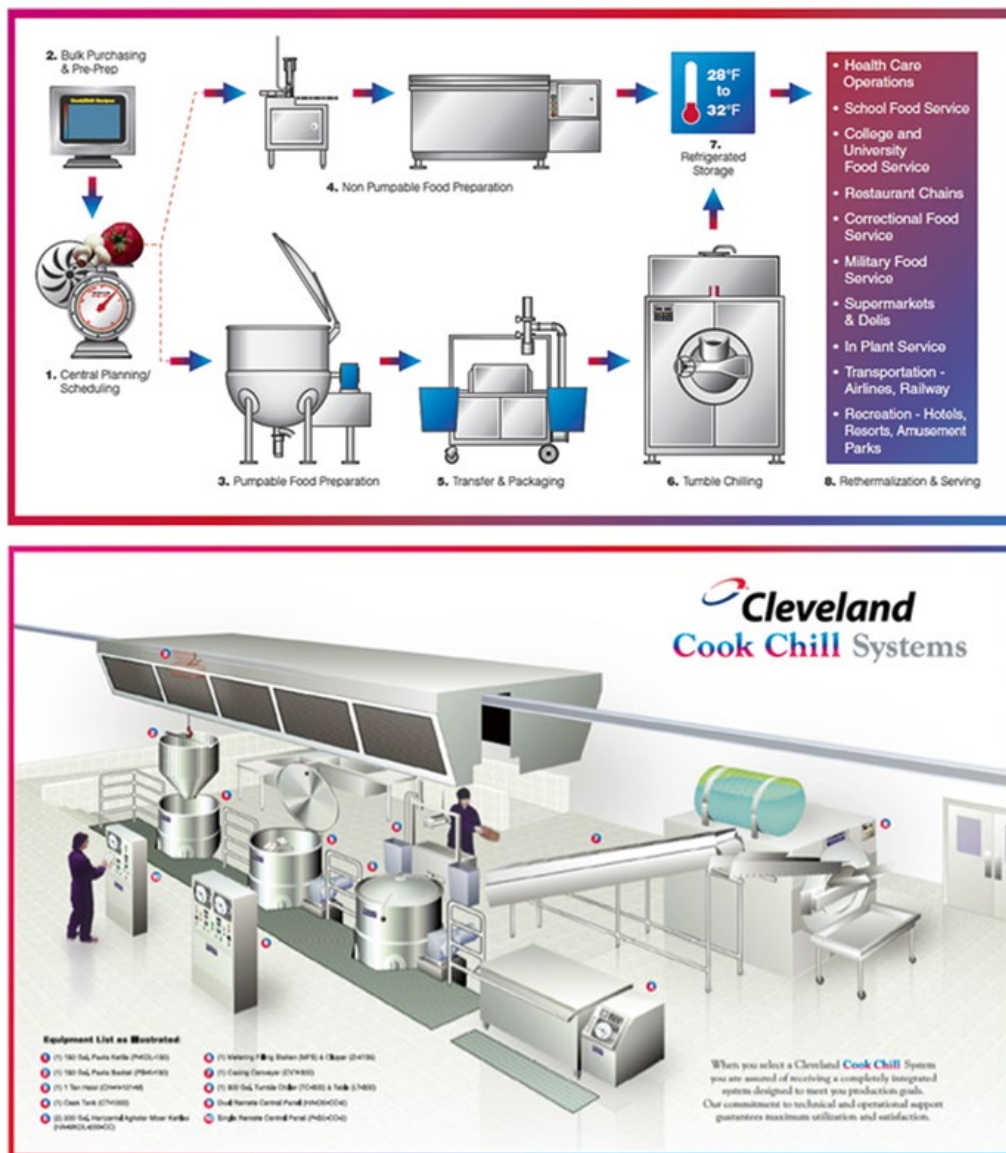


Fig. 7.3 Production line for cook-chill foods

Figure 7.4 shows a processing scheme of cook-chill/freeze products. The technology is a process that consists of blast freezing and frozen storage followed by thawing and then display at chill conditions (O'Leary et al. 2000; Zhang and Sun 2006).

7.6.1 Processing Conditions

Processing conditions require appropriate temperature and time and vacuum packaging (del Pulgar et al. 2013). Since the safety, nutritional value and sensory quality are the main concerns for cook-chill meals, it is essential to understand how

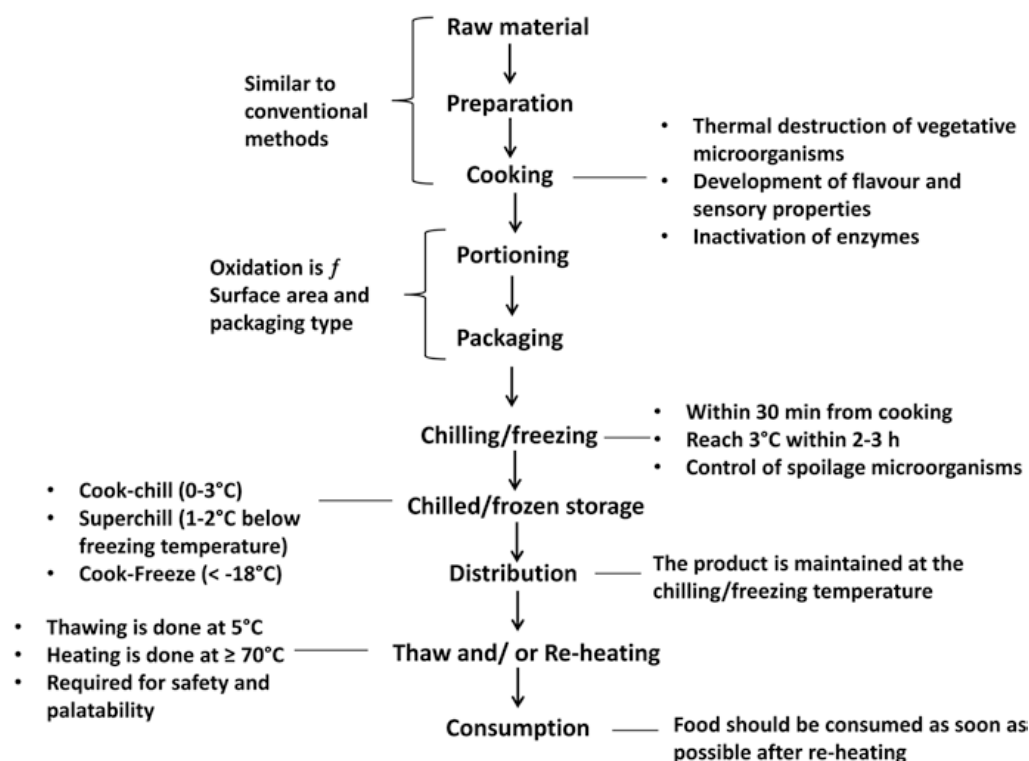


Fig. 7.4 Flow chart for different operations of cook/chill and cook-freeze foods

the cooking method (temperature and time) and processing technique (e.g. atmosphere, vacuum, and high pressure) affect quality. The preparation steps (i.e. including heating and cooling) are crucial factors for retaining nutritional value and sensory quality. High cooking temperatures or longer cooking times at moderate temperatures can deteriorate the appearance and texture of vegetables, e.g. degradation of chlorophyll in green beans during cooking (Stolle-Smits et al. 1997; Van Boekel 1999). Abuse of temperature during cooking, cooling or storage can promote further degradation of the sensory attributes (Krebbbers et al. 2002; Martins et al. 2004). Therefore, temperature control is very important for achieving better sensory properties.

7.6.2 Production of Cook-Chill Products

7.6.2.1 Raw Materials and Ingredients

Various control measures must be implemented in order to reduce or eliminate contaminations of cook-chill foods at the initial stages of processing. Appropriate selection of raw materials and ingredients is the first step to achieve a high quality product. Control measures to produce consistent quality and safe products include washing, heating, disinfecting, sorting, grading, transportation and cold storage.

245 Applying quality control plans and good manufacturing practice to eliminate hazards
246 and possible contamination can be integrated easily in cook-chill/cook-freeze
247 production systems.

248 **7.6.2.2 Heating Techniques**

249 The most common systems used for cooking/heating are water bath or steam. However,
250 conventional heating systems can produce an inconsistent heating profile, especially if
251 large volumes of food are processed. Sheard and Rodger (1995) examined ten heating
252 systems and reported that none of the convective steam ovens heated sous-vide pouches
253 uniformly under fully loaded operational conditions. It took 16–85.5 min to increase
254 the temperature of sous-vide pouches from 25 to 75 °C. The variations depend on the
255 location of the product within the oven as well as the types of oven. The circulating
256 water baths provided uniform heating with a variation of 0.05 °C. It is important to
257 submerge the pouches (in a rack) during cooking to avoid undercooking or potential
258 microbial growth during post-processing (Rybka-Rodgers 1999).

259 **7.6.2.3 Rapid Chilling or Freezing Equipment**

260 The refrigeration systems used for REFPFED are: blast chillers, blast freezers or
261 combination of both. These are capable of rapid reduction of temperature of hot
262 foods (+70 °C) to low safe storage temperatures (+3 °C or –18 °C). The basic prin-
263 ciple of food safety in cook-chill food products is the start of cooling as soon as
264 cooking is finished and the final core temperature of the product should be below
265 4 °C. This can be achieved using specially designed rapid chillers, such as air-blast
266 chillers and vacuum coolers. These are capable of decreasing the temperature of the
267 hot food to 0 °C or +3 °C within 90 min. This ensures the product safety as well as
268 preserving the sensory attributes. As the final cooling temperature and the rate of
269 cooling are crucial control points in this process, chillers are normally equipped
270 with temperature probes at various locations to monitor the process.

271 There are two main methods of cooling used for cook-chill and sous-vide prod-
272 ucts: blast chilling and water bath chilling. In blast chilling, cold air circulates over
273 food containers (usually shallow pans), while in the water bath method, cooked
274 food containers are submersed in cold water. Rapid cooling or freezing (also known
275 as hard blast chilling) is used for dense, large products with high fat content (e.g.
276 meat based sauces, meat joints, mashed potato and lasagna). The air temperature of
277 the cabinet, normally below the freezing temperature of the product, must ensure
278 that the core of the product reaches the required +3 °C within the 90 min. A slower
279 chilling rate is obtained by “soft blast chilling”. This technique uses air temperature
280 above 0 °C in order to avoid any damage to delicate and light products, such as fish,
281 rice, vegetables, cream, desserts, cakes and fried foods. Therefore, the soft blast
282 chilling cycle smoothly lowers the product core temperature to +3 °C over 90 min
283 without compromising the quality of delicate products.

For cook-freeze processing, the freezing must commence within 30 min of cooking and must achieve a core temperature of at least -5°C within 90 min. The final temperature should be -20°C or less. The cooling rates (from around 60°C to $+3^{\circ}\text{C}$ within 90 min) require very powerful refrigeration equipment that blows high velocity chilled air into the refrigerator to achieve cooling at high speed. The heat transfer increases with increasing air velocity (decreasing the boundary layer on the surface of the product) and temperature difference between the air and the product. In the case of direct contact with food products, stainless steel needs to be used for maintaining hygiene and easy cleaning. Hot food in bun pans or any other containers should be well spaced during blast chilling. The use of shallow containers is preferred since they can cool fast by minimizing the heat transfer resistance. The hot food should be uncovered so that an accelerated cooling rate can be achieved. This method is recommended for food that will be consumed within a few days (i.e. up to 5 days; Garrett 2012).

Another method used for rapid chilling is tumble chilling. After cooking and filling the product into casings, the product is loaded onto a rotary stainless steel chiller, which reduces the temperature of the cooked food below 4°C within an hour (i.e. foods with low viscosity) or within 90 min for viscous foods. The bagged product is loaded into a drum which rotates in a chilled water tank (i.e. 400–4,000 l capacity). The water around the product is chilled by circulating a chilled refrigerant (e.g. propylene glycol) through a dimpled jacket in the outer casing of the unit. The primary cooling medium may be any refrigerant provided that heat exchange between both the refrigerant and the food is properly installed. A tumble chiller can hold from 25 to 450 kg of product in one cycle. This method is widely used and it results in consistent food quality with lower costs associated with energy and labour (Vomvoris 2005).

[AU1] Another chilling system used is vacuum cooling. It has been traditionally used as a precooling technique for fresh leafy vegetables, such as spinach and lettuce (Elansari 2008). Several studies documented the use of vacuum cooling as an efficient technology to chill foods, such as cooked rice, ready meals and cooked meats (Drummond and Sun 2008). Vacuum cooling of cooked meat was investigated by Zheng and Sun (2004). Vacuum cooling can significantly increase the cooling rate of cooked meat. Vacuum cooling reduced the temperature of cooked meat from 70 to 74 to 4°C in 1–2.5 h compared to air blast chilling, whereas slow air chilling, or water immersion chilling, took 9.4–11.7, 12.1–14.3 or 5–14.3 h, respectively. Zhang and Sun (2006) also reported superior processing efficiency (shorter cooling times, lower evaporative losses from product, and less water-spraying used) of vacuum cooling compared to the cold room, blast cooling and plate cooling of broccoli and carrot slices. The higher cooling rates obtained via vacuum cooling compared to other conventional cooling methods are due to differences in cooling mechanisms (Tragethon 2011). For conventional cooling methods, heat is mainly removed by an internal conductive heat transfer from the core of the product to the surface. The surface heat is then removed by convective heat transfer in cooled air or water. The internal heat transfer by conduction is slow due to the low thermal conductivity of cooked meat, and therefore it becomes the rate-limiting step. This explains why the

cooling rate of conventional systems is largely dependent on sample size (i.e. faster cooling time is obtained with small size samples; McDonald et al. 2000).

On the other hand, vacuum cooling takes place by evaporating water from the surface or pores inside the product at low atmospheric pressure (Kader 2002). Water evaporation takes place on the surface of the macro-pores of the materials followed by diffusion of water vapour through the pore spaces to the surface and subsequently water is lost to the surrounding atmosphere (Wang and Sun 2002). Therefore, the porous structure of the samples (e.g. pore size, pore shape, total pore spaces, pore distribution) plays a critical role in determining the water evaporation rate and subsequent vacuum cooling rate. Products that easily release water may cool to the desired chilling temperature in 20–30 min. Vegetables with high surface:mass ratio release water rapidly (e.g. leafy green vegetables, celery, sweet corn, green beans, carrots, and bell peppers) and these are best suited for this method. Boxes of film-wrapped cook-chill products can cool quickly provided the film allows easy movement of water vapor.

The operating temperatures in cook-chill and freeze-chill steps must therefore be sufficiently low to inhibit growth of the relevant microorganisms (Baerdemaeker and Nicolai 1995). As mentioned earlier, the processing conditions applied (chilling rate and cold storage temperature) vary among different EC countries (James 1990). In France, the recommended temperature needs to be 10 °C, decreasing from 70 °C within 2 h, followed by storage within 0–3 °C. In other countries such as Denmark, Germany, Sweden and UK, the chilling times are 3 h (i.e. from 65 to 10 °C), 2 h (i.e. from 80 to 15 °C), 4 h (i.e. from 80 to 8 °C) and 1.5 h (i.e. from 70 to 3 °C), respectively (Hill 1994). It should be noted that these recommended times depend on the package size. For example, the guidelines of the Department of Health in the United Kingdom for cook-chill products are that 80-mm trays should be chilled below 10 °C in 2.5 h and the smaller 40- and 10-mm trays should be chilled below 3 °C in 1.5 h. These recommendations were verified using Bolognese sauce in trays with various sizes (Evans et al. 1996).

Inappropriate vacuum during processing leads to the formation of insulating air layers between product and package. This represents an additional resistance to heat transfer causing a delay in the cooling cycle and needs to be accounted in refrigeration load calculations. Other factors that need to be considered in the design of chilling processes are; (1) size, shape, weight of food and container; (2) material of the container; (3) density of food and its moisture content; (4) heat capacity of the food and the container; (5) thermal conductivity of the food; (6) design of the chiller, which affects chilling speed; (7) temperature of the food entering the chiller; and (8) lid or cover of the container (Hill 1994).

7.6.2.4 Packaging

The application of cook-chill packaged products in the food service industry can provide a buffer zone and flexibility in meal preparation. Packaging of sous-vide products provides an example of how packaging can significantly improve the

storage stability of products (Jang et al. 2006; Leistner and Gould 2002). Many products prepared by conventional and labour-intensive methods are now processed by cook-chill (especially sous-vide since 1980s). This is due to the perceived superior sensory qualities as well as the microbiological stability (Creed 1998; Werlein 1998). The application of sous-vide for traditional products are gaining lots of interest (Schafheitle and Light 1989; Simpson et al. 1994; Kim et al. 2001).

The sealed laminated-plastic pouch or containers under vacuum are used for sous-vide foods. These containers contain low oxygen partial pressure and prevent the growth of aerobic spoilage micro-organisms and lipid oxidation causing rancidity. Lipid oxidation generates off-flavours during refrigerated storage (Church and Parsons 2000). Vacuum pouches should provide good mechanical resistance, thermal tolerance (-40 to $+120$ °C) and provide excellent heat transfer capabilities. Vacuum packing also helps to maintain the product quality and to extend the product's shelf life by three- to fivefold.

For sous-vide, packs are typically brick-shaped with a thickness of 3–5 cm, whereas ready-meals/cook-chill bags are used with a wide range of shapes. The pressure inside the bag is preferably below 1.5 kPa for meat and vegetables (Rozier et al. 1990). In the case of liquid foods (e.g. sauces and soups) the inside pressure limit is 10 kPa (Baerdemaeker and Nicolai 1995) since it is very difficult to achieve low pressure.

In the case of sous-vide, it is common to have inflated packages during re-heating if there is an insufficient vacuum (Baerdemaeker and Nicolai 1995). Therefore, this effect needs to be considered for transfer calculations. Meals accompanied by sauces or other liquids represent a significant challenge for cook-chill/sous-vide processing due to their probability of rupturing during processing, handling and storage (Díaz et al. 2010). A recent trend to alleviate this problem is the use of plastic trays sealed under low oxygen conditions and the use of steam permeable plastic films. The plastic trays are commonly accompanied by gas mixture flushing. This can prevent failure of the packages, avoid mechanical damage, and eliminate head-space oxygen. A main disadvantage of this type of packaging is the formation of condensed moisture in the cooked product. However, these plastic trays are more robust, easier to fill (i.e. especially in the case of liquids), more attractive to consumers and the trays can be re-heated before consumption (Díaz et al. 2010). Thus it makes a very attractive option for the industry.

It was found that MAP conditions of 60 % CO_2 /40 % N_2 extended the shelf life of prepared meals stored at 10 °C by 233 % (to 18.4 days) compared to the stretch-wrap air packaging (7.9 days; Lee et al. 2008). The shelf-life was determined based on the time required to grow anaerobic bacteria at 10^5 CFU/g. Moreover, the relative extension of the shelf life with MAP was greater under lower temperature storage. MAP was very effective in preserving the sensory properties and microbiological quality of a cook-chilled Korean traditional rice cake (backseolgi) stored at 3 °C (Lee et al. 2011). After 50 days of storage, MAP samples showed 1.0 log CFU/g, while low density polyethylene showed more than 5.0 logs CFU/g after 4 days.

Levkane et al. (2010) investigated the effect of the packaging technology on the microbial growth of sous-vide salad containing meat in mayonnaise. The product was either packed in polystyrene (PS) containers (of size 100 mm $\times$ 80 mm $\times$ 25 mm,

and polystyrene thickness $30 \pm 3 \mu\text{m}$) covered with non-hermetical lids, or sous-vide processed in polyamide/polyethylene (PA/PE) film pouches with barrier properties (of size $200 \text{ mm} \times 300 \text{ mm}$, and PA/PE film thickness of $20/45 \mu\text{m}$). The products were chilled at either 4 or 10 °C for up to 52 days. Control samples were microbiologically spoiled after 10 days of cold storage at 4 °C. Sous-vide processing significantly reduced microbial growth at 4 or 10 °C and the product shelf life was extended to 52 days even at a chilling temperature of 10 °C. The benefits of packaging in the case of cook-chill under vacuum can be summarized as follows:

1. It can maintain delicate flavours (i.e. all the natural goodness and taste of food can be preserved for 4–6 days in the fridge or frozen storage for much longer periods, up to 2 years) (Newton and Gillespie 2010). There are negligible studies available in the literature on the effect of vacuum on the volatile profile of sous-vide cooked meat at low to moderate temperatures.
2. It can inhibit oxidation. Vacuum packing helps to preserve cook-chill food by removing oxygen-containing air inside the bag and thereby limiting deteriorative oxidative processes.
3. Vacuum packaging of food eliminates freezer burn by protecting the food from evaporation during frozen storage. Water loss from the surface can cause leathery appearance, discoloration, development of off-flavor and unacceptable texture. In a similar way, air circulation inside freezing rooms or cold stores causes moisture loss by evaporation from the product surfaces. Minimizing water loss can keep food moist and juicy.

7.6.2.5 Process Validation and Safety

Temperature control during heating, cooling and storage of cook-chill foods is very important. It has been reported that oven and cold store temperature measurements at selected central locations were not satisfactory (Baerdemaeker and Nicolai 1995; Schellekens and Martens 1992). While non-homogenous spatial temperature distribution in the various heating or cooling systems can be expected, it is important to consider this point to calculate lethality and cooling rates. The overall effectiveness of the control should be validated according to the prevalence of microbial hazards, desired food safety levels, and perceived risk (Zwietering et al. 2010; CAC 2008).

The validation of cook-chill processes needs to consider the following steps: cooking, cooling, cold storage, transport to the catering facilities and reheating before serving (Poumeyrol et al. 2012). The most significant factor likely to lead to *Clostridium perfringens* outbreaks in cook-chill food is the cooling step (Golden et al. 2009; De Jong et al. 2005; Poumeyrol et al. 2012). The cooling step is often poorly controlled and not highly efficient in the catering sector (Olds et al. 2006). This is due to inappropriate equipment design and inadequate cooling capacity. This slow-cooling results in the germination of spores and growth of vegetative cells. To counter this hazard, guidelines and codes for cook-chill production have been established in different countries. These guidelines can avoid temperature abuse during the cooling step. The French public health authorities enforce a defined

temperature-time combination, from 63 to 10 °C within 2 h (French Ministry for Food, Agriculture and Fisheries 2003), while the USDA recommendation (2001) consists of two values, namely a cooling from 54 to 26 °C in less than 90 min and from 26 to 4 °C in less than 5 h. For the Australian and New Zealand Food Standards Code (NSW 2011), all cook-chill foods must be cooled after cooking in accordance with Food Standards Code requirements (the '2-h/4-h cooling rule'): (a) within 2 h—from 60 to 21 °C, and (b) within a further 4 h, from 21 to 5 °C, unless the processor can demonstrate that the cooling process used by them will not adversely affect the food safety.

Several cases of temperature abuse have been reported in the literature (Jaloustre 2011; Poumeyrol et al. 2012). For example, there are cases when the product was removed from the blast-chillers at a temperature above 10 °C, but this temperature was not representative of the bulk average temperature of the whole batch (i.e. some packs were substantially higher than 10 °C). This can potentially cause problems with shelf life and product safety. In addition, effective systems of temperature measurement are needed to be in place to enable validation of cooling processes.

7.6.2.6 Freezing as an Alternative

If temperature abuse occurs with sous-vide and cook-chill foods, these may become a source of health risk. This is due to the potential anaerobic growth of *C. botulinum* spores in the food (Gould 1999; Gaze and Brown 1990; Bolton et al. 2000; Tansey et al. 2010). The risk of temperature abuse is greater during different stages of the food handling process, such as during transport and storage at the final destination (i.e. home, hotel or any other catering facility). Therefore, frozen sous-vide or a cooked food is regarded as a safer alternative to sous-vide-chilling and cook-chill processing. As mentioned earlier, cook-freeze process is the same as the cook-chill process, except the cook-freeze product is frozen to −20 °C in the central processing facility. In the catering sector, cook-freeze food can be stored for up to 8 weeks before it is reheated for consumption. The potential growth of *C. botulinum* spores are eliminated by freezing (Tansey et al. 2010). Freeze-chill technology involves initial freezing and frozen storage (−30 °C or colder), followed by thawing and chilled retail display usually at 4 °C (O'Leary et al. 2000; Redmond et al. 2004). Freeze-chill technology allows 'chilled' foods to reach more distant markets, facilitates bulk production, and reduces the level of product recalls.

Cook-freeze and freeze-chill processing may not be suitable for all food products and can adversely affect the quality attributes of some food items. For example, it was reported that freeze-chilling leads to a reduction in the vitamin C content of mashed potato (Redmond et al. 2004). In addition, freezing affects texture due to structural damage from ice crystal formation. Depending on the freezing rate, several physical changes are associated with the formation of large sharp edged ice crystals, combined with volume expansion, and disruption of the osmotic equilibrium between the cells and their surroundings. These may cause irreversible damage to the texture of vegetables, fruits and meats. The ice crystals formed during slow

freezing of vegetables leads to loss in cell turgor pressure, weakening the cell wall, increased permeability across cell walls, and consequently a softer texture and higher drip loss on thawing (Fuchigami et al. 1995; Préstamo et al. 1998). Fast freezing rates result in the formation of a larger number of ice nuclei and consequently smaller ice crystals and firmer texture (Fuchigami et al. 1995; Rahman et al. 1971). Frozen storage temperature also influences the texture of vegetables. The lower frozen storage temperatures result in slower growth in the size of ice crystals and may prevent re-crystallization, and consequently yield optimum texture preservation (Kennedy 2003).

7.6.2.7 Thawing and/or Reheating and Serving

Cook-freeze and freeze-chill technologies require thawing of the frozen food prior to retail display and consumption. It is essential that the system design and selection allow thawing and all ice crystals are entirely thawed before the food is re-cooked. Any remaining unthawed parts upon subsequent re-heating may not achieve adequate safe and well-distributed temperature environments, which can present an increased risk of food poisoning (Barrie 1996; Brown et al. 2006).

Generally, thawing occurs more slowly than freezing and foods are subjected to more damage during thawing (Li and Sun 2002). The difference between freezing rate verses thawing rate are due to differences in thermal conductivity and diffusivity of water and ice (Park et al. 2006). In addition, the temperature of foods increases rapidly just before freezing, while there is a long lag time for phase transition at the freezing and melting plateau leading to tissue damage due to re-crystallisation, microbial activity and physicochemical changes (Fennema et al. 1973). Therefore, optimum thawing procedures should be implemented for cook-freeze foods (Fennema et al. 1973; Kalichevsky et al. 1995).

During thawing, temperatures must not rise above 10 °C according to the UK guidelines (Anon 1989), but a stricter thawing practice is recommended in Australia (i.e. no more than 5 °C; NSW Food Authority 2011). Following thawing, it is recommended that the temperature of the food must be reduced to below 3 °C homogeneously throughout the food within 12 h. In practice, this means that any thawing process greater than 3 °C must be followed by a temperature reduction stage, termed re-cooling (Brown et al. 2006).

Thawing of cook-freeze foods should be done under conditions that minimize microbial growth. Thawing methods used in the food industry vary in terms of thawing media as well as the nature of the heating applied. The thawing methods include the use of hot air, immersion in warm liquids, steam at ambient pressure or under reduced pressure. These reduce surface temperature gains, while higher surface temperatures are observed during microwave or direct contact heating plates (Koke et al. 2002). New methods have been developed more recently, including high-pressure and acoustic thawing (Li and Sun 2002). Thawing at ambient temperature can lead to the uncontrolled and unrecognized growth of pathogens and

should only be applied when risks have been properly assessed. A controlled thawing cabinet with hot air is the most common applied method for this purpose. However, microwaves or steam are not practically suitable for cook-freeze products, since they cause undesirable surface conditions. Thawing using a microwave oven may leave cold spots at the core of the food unless the microwave is specifically designed to ensure even thawing.

Thawing is a critical process and must be considered within a HACCP plan. Hazards associated with thawing include cross-contamination from the drip and growth of micro-organisms on the outside of the package before inside thawing. Thawed meat and poultry products should be checked frequently to make sure of a complete thawing process before further processing. As a general rule, rapidly thawed foods in fast thawing cabinets should be consumed within 24 h. Thawed foods should not be re-frozen due to quality and safety issues.

Similar to the freezing rate prediction models of foods, several thermodynamic models of the thawing process have been reported in the literature (Cleland et al. 1986; Salvadori and Mascheroni 1991; Ramos et al. 1994; Ilicali et al. 1999). The key feature of these models is the analytical equations for predicting a thawing time, which relates to the melting of all ice in the product. However, temperature time history of the product should also modeled for complete thawing. Therefore, data on thawing times for foods in cook-freeze packs are essential for correct design and operation of these systems. Computer modelling was used to identify the fastest and slowest thawing packs from a typical hospital catering range (Brown, et al. 2006). It is important that practical systems are able to achieve acceptable uniformity of air temperature and velocity over each and every pack, otherwise thawing times can vary greatly. This requires careful design of air flow and pack spacing, otherwise it would be necessary to allow considerable tolerances in thawing times.

7.6.2.8 Reheating

Cook-chill products are fully cooked and stored chilled until their consumption. At the time of use/consumption the food is often reheated to achieve: (1) improved product aroma and flavour resulting from the generated volatiles during the heating step, and improved palatability due to changes in the perceived texture, and (2) improved food safety by killing any microorganisms that may be present in the stored product.

Reheating (also called re-thermalization) should be initiated before food consumption (i.e. within 30 min from the removal of the food from chilled storage). Reheating can be carried out using microwaves, forced air and steam convection ovens. Traditional hot-air ovens can potentially cause drying of exposed areas of the food. Commercial microwave ovens are very efficient in terms of time and energy use, but are recommended for the reheating of individual portions or small numbers of meals. Other options of reheating methods include: infrared ovens, hot water baths, convection steamers, combination steamer/ovens, and hot plates (Yang 1990).

583 Reheating foods by one of the above mentioned methods provides an efficient
584 means to prepare the number of meals required at a given time, thus minimizing
585 food wastage.

586 The use of retherm carts to reheat meals is a more recent development in hospital
587 food services. The retherm cart electric heating discs are used to warm meals in a
588 refrigerated room, while other chilled products such as fruits and drinks, and remain
589 chilled until service. This system can heat between 12 and 26 meals on metal trays.
590 Reheating of cook-chill food products should be strictly carried out by trained per-
591 sonnel under strict conditions. This step can significantly impact the nutritional
592 quality of food, for example significant losses of thiamin and ascorbic acid are
593 reported to occur during long heating periods (Edwards and Hartwell 2006;
594 McErlain et al. 2001).

595 **7.6.2.9 Facility Design and Equipment**

596 The ultimate objective of cook-chill process design is to automate and control the
597 stages of the prepared food manufacturing so that it can occur as a steady and
598 smooth process, rather more typical peak-and-valley method of most kitchens. Food
599 quality can be fine-tuned in cook-chill processing through strict adherence to stan-
600 dardized recipes and procedures to ensure a sustainable and consistent product. It is
601 important to design processing facility solely for a particular food preparation pro-
602 cess (i.e. cook-chill). Conventional walk-in refrigeration units cannot reduce food
603 temperatures quickly, and therefore should not be an alternative for the air blast
604 freezer. The equipment needs automatic transfer, portion control, and packaging of
605 finished cook-chill foods. The layout and design of the facility, equipment and
606 maintenance need to be designed properly.

607 The layout and design of the facility should allow an efficient flow of work that
608 facilitates movement of both products and personnel through the facility. The food
609 should ideally move the shortest distance possible with reduced crossing paths as
610 much as possible. One example of a food workflow is shown in the flow chart in
611 Fig. 7.5. The layout should also maximize the productivity of employees and take
612 into consideration the possibility of future expansion. All utilities such as water sup-
613 ply, drainage, electricity and lighting should be provided at the right locations with
614 good safety and sanitary considerations. The total amount of space required for the
615 facility needs to adequately accommodate all equipment and processes. Appropriate
616 space is needed for clearways to satisfy the movement of cart, forklift and the pallet
617 jacks. Adequate space is required for accessing equipment for maintenance and
618 cleaning. The main functional facilities should be properly laid out, include areas
619 for receiving and storage of raw product, hot food preparation, cold food prepara-
620 tion, bakery, packaging, assembly, storage of prepared foods, shipping/distribution,
621 and equipment cleaning.

622 The type of equipment, size and their sequence influences the layout and design
623 of the facility of the production. Moreover, the required equipment layout depends

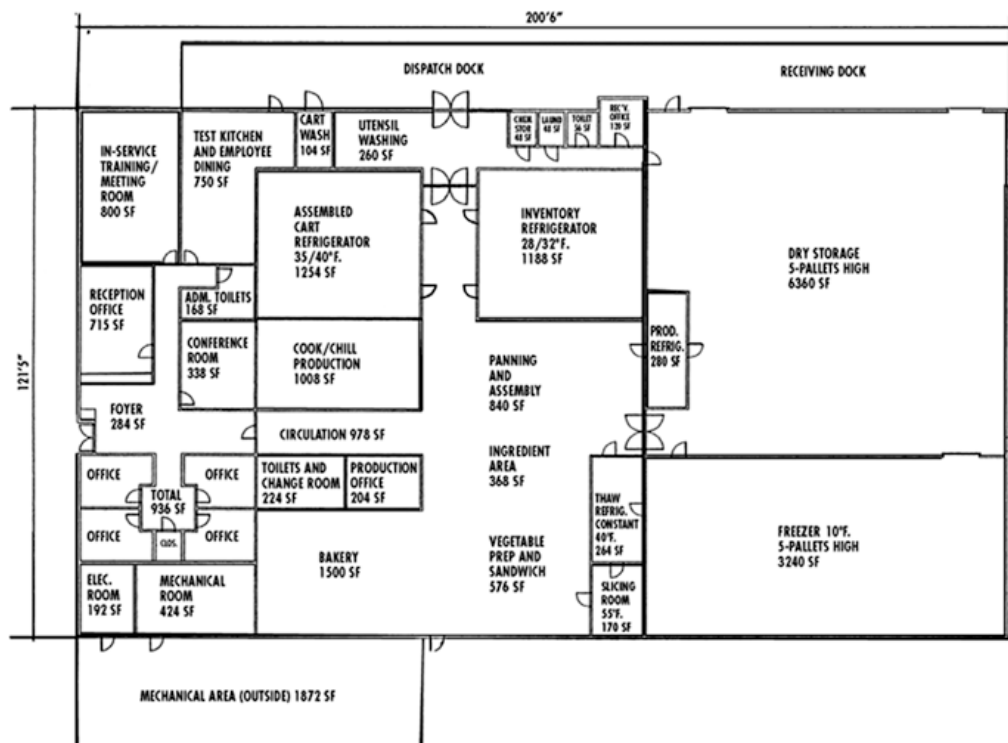


Fig. 7.5 Layout and design of a food processing facility

on the desired items and the functional areas used. Equipment sizing and selection should be based on detailed technical specifications and it should provide fine control, robust, hygiene, durability, less maintenance and power saving options and capabilities. The specifications do not only describe equipment, but also include information about electrical requirements, plumbing requirements, steam requirements, and mechanical requirements. The specifications should also include information concerning freight, delivery and installation requirements, handing over conditions, warranty and extended warranties, instructions for installation and removal of old equipment and training required (NFSMI 2002).

A progressive maintenance plan (preventive and predictive) for equipment must be developed to ensure that the facility can meet the sustainable production objectives and minimize any delay caused by an emergency breakdown. The impact of an equipment breakdown in a facility would have a significant cost on the production and interrupt the scheduled operations. Thus, detailed and supported plans must be in place for both preventive and predictive maintenance along with an inventory of spare parts needed and trained persons. A health and safety emergency plan should be developed to guide workers in certain circumstances, such as a power outage. Other items should be considered include design of the dryer, chilling and freezing warehouses and careful planning for it. In addition, a waste management system should be in place based on the rules of the local authority.

644 7.6.2.10 Thermal Properties and Prediction Models

645 The thermophysical properties of foods depends on the heat and mass transfer
 646 mechanisms, phase changes involved, and chemical reactions during chilling, freezing
 647 or heating. These characteristics include temperature and velocity, fluid properties
 648 such as viscosity, density, conductivity, and heat capacity, product surface proper-
 649 ties such as geometry and dimension, and internal structure or arrangements. The
 650 dielectric properties of foods are very important for microwave heating. Due to the
 651 interaction of these parameters, the boundary layers can be very complex, therefore
 652 predicting the cooling time by considering all the above factors can be a difficult
 653 task (Hu and Sun 1999, 2000).

654 Convection heat transfer is the major mode of heat transfer between the surface
 655 of a solid material and the surrounding fluid. The rate of convective heat transfer
 656 depends on the properties of the fluid and the fluid flow characteristics. Originally
 657 as suggested by Prandtl, the resistance to heat transfer may be considered to be
 658 localized in a boundary layer within the fluid present at the surface of the solid
 659 material. Although this concept is for ideal situations, it has been widely used in
 660 studying convective heat transfer. Commercial numerical techniques, such as com-
 661 putational fluid dynamics (CFD) using CFX software have been successfully
 662 adapted to the food industry to simulate the thermal process. This can provide a
 663 better understanding of the mechanism and aid in the improvement of design and
 664 operation (Anon 1997). Recently, a number of models have been developed for
 665 predicting the food chilling process and the influence of its thermal properties
 666 (Chuntranuluck et al. 1998; Kuitche and Daudin 1996; Davey and Pham 1997;
 667 Maroulis et al. 1995; Hu and Sun 1999, 2000; Xia and Sun 2002; Kaushal and
 668 Sharma 2012).

669 Computational fluid dynamics can be used for other equipment designs, for
 670 detailed product development and for scale-up of the process. The microbiological
 671 stability and safety of sous-vide products is mainly determined by the different ther-
 672 mal treatments including heating, cooling and reheating. The theoretical aspects of
 673 the different modes of heat transfer relevant to 'sous-vide' cooking can be opti-
 674 mized using computational techniques such as the finite element method and
 675 CFD. These techniques can provide a powerful means to investigate the effect of
 676 modifications in the processing conditions on the internal temperature of the food
 677 (Baerdemaeker and Nicolai 1995).

678 A CFD model to study heat and mass transfer of a cylindrical shaped cooked
 679 meat within an air-blast chiller was carried out to predict its cooling rate and weight
 680 loss during chilling (Hu and Sun 2000). The investigation was based on a mathe-
 681 matical analytical model of unsteady heat and mass transfer with the assumption of
 682 a homogeneous heat transfer coefficient (i.e. which takes into account the effects of
 683 forced convection, radiation and moisture evaporation on the surface of the cooked
 684 meat joint). This method allows the simultaneous CFD prediction of both tempera-
 685 ture distribution and weight loss in the meat throughout the chilling process. Cooling
 686 time and weight loss from 75 to 3.5 °C were approximately 530 min and 4.25 %,

respectively, by both experiment and prediction model. The simulation using CFD
can be easily adapted to other elementary shapes, operating conditions of the cook-
chill production chain.

7.6.2.11 Effect on Texture and Nutrition

Cooking can have a dramatic effect on the texture of food depending on the cooking
temperature, food material structure/composition/condition and method of cooking.
The cook-chill process produced more soft cooked dry peas than sous-vide cooking
due to the higher temperature of cook-chill processing. This improvement is due to
moisture uptake (Xie 2000). MAP cook-chilled Korean traditional rice cakes scored
significantly higher than LDPE packaging after 50 days of storage at 3 °C (Lee et al.
2011). Sous-vide processing reduced stem broccoli texture (softening %) by
54–58 % (i.e. a pleasant and moderate softening level). It was one of the treatments
(as well as grilling and steaming compared to vacuum boiling, conventional boiling,
pressure cooking, and microwaving) that induced the lowest stem colour changes
(Martínez-Hernández et al. 2013).

While sous-vide processing decreased the total folate contents in potato and
broccoli by 20.9 and 16.5 %, respectively, total folate content was increased by
4.9 % in sous-vide processed, cooled and reheated potato (Stea et al. 2006).
However, other ingredients are used during the assembly of the meals, thus new
nutrient profiles may arise. Shakila et al. (2012) reported the reduction of saturated
fatty acids and an increase in the polyunsaturated fatty acids in the case of sous-vide
processing of fish curry. While the observed changes are likely to be related to the
inclusion of other ingredients, it demonstrated the ability of the technology to
manipulate the nutritional composition favourably. The total antioxidant capacity of
broccoli was increased by sous-vide, microwaving and frying compared to steam-
ing, pressure cooking, and conventional boiling as well as in the raw state (Martínez-
Hernández et al. 2013).

7.7 Conclusion

Refrigerated cook-chill foods deliver convenience and safety with retention of
nutritional content. The quickly prepared meals are needed for families and indi-
viduals depending on their life style. The good safety track record of cook-freeze
and sous-vide food products has created a trust. These are considered very accept-
able and are seen as a better alternative for branded fast food. Appropriate process-
ing conditions for microbiological safe cook-chill products are relatively established
for many products. Challenges that require future research are: (1) optimizing the
sensory quality of the food, and (2) developing a wide range of menus so that these
can match cook-serve options. These products are appealing, however strict han-
dling conditions need to be implemented.

725 **References**

- 726 Anon (1989) Chilled and frozen guidelines on cook-chill and cook freeze catering systems. UK
727 Department of Health. ISBN 0-113211- 61-9
- 728 Anon (1997) Guidelines for good hygienic practices in the manufacture of chilled foods. Chilled
729 Food Association, London
- 730 NSW (Food Authority, The Australia New Zealand Food Standards Code) (2011) Guidelines for
731 food service to vulnerable persons. NSW Food Authority 6 Avenue of the Americas Newington
732 NSW 2127
- 733 Baerdemaeker JD, Nicolai BM (1995) Equipment considerations for sous vide cooking. Food
734 Control 6:229–236
- 735 Baldwin DE (2012) Sous-vide cooking: a review. Int J Gastron Food Sci 1:15–30
- 736 Barrie D (1996) Infection control in practice. The provision of food and catering services in hospital.
737 J Hosp Infect 33:13–33
- 738 Bolton DJ, McMahon CM, Doherty AM, Sheridan JJ, McDowell DA, Blair IS, Harrington D
739 (2000) Thermal inactivation of *Listeria monocytogenes* and *Yersinia enterocolitica* in minced
740 beef under laboratory conditions and in sous-vide prepared minced and solid beef cooked in
741 commercial retort. J Appl Microbiol 88:626–632
- 742 Brown T, Evans JA, James C, James JS, Swain MVL (2006) Thawing of cook-freeze catering
743 packs. J Food Eng 74:70–77
- 744 CAC (Codex Alimentarius Commission) (2008) Guideline for the validation of food safety 499
745 control measures. CAC/GL 69. FAO, Rome
- 746 Carlin F, Guinebretiere MH, Choma C, Schmott P, Nguyen-The C (1999) Spore-forming bacteria
747 in cooked chilled foods containing vegetables. In: Proceedings of the 3rd European symposium
748 on Sous Vide, Leven, Belgium, Alma Sous Vide Competence Centre, pp 55–67
- 749 Chater A (2000) Making it work for you: the cook-chill craze. Foodserv Dir 13:140, <http://search.ebscohost.com/login.aspx?direct=true&db=heh&AN=3831946&site=ehost-live&scope=site>
- 750 Chudasama Y, Hamilton-Miller JMT, Maple PAC (1991) Bacteriological safety of cook-chill food
751 at the Royal Free Hospital, with particular reference to *Listeria*. J Hosp Infect 19:225–230
- 752 Chuntranuluck S, Wells CM, Cleland AC (1998) Prediction of chilling times of foods in situa-
753 tions where evaporative cooling is significant—Part 1. Methods development. J Food Eng
754 37:111–125
- 755 Church IJ, Parsons AL (2000) The sensory quality of chicken and potato products prepared using
756 cook-chill and sous vide methods. Int J Food Sci Technol 35:155–162
- 757 Cleland DJ, Cleland AC, Earle RL, Byrne SJ (1986) Prediction of thawing times for foods of
758 simple shapes. Int J Refrig 9:220–228
- 759 Creed PG (1998) Sensory and nutritional aspects of sous vide processed foods. In: Ghazala S (ed)
760 Sous vide and cook-chill processing for the food industry. Aspen Publishers, Gaithersburg, MD,
761 pp 57–88
- 762 Davey LM, Pham QT (1997) Predicting the dynamic product heat load and weight loss during beef
763 chilling using a multi-region finite difference approach. Int J Refrig 20:470–482
- 764 De Jong AEI, Beumer RR, Zwietering MH (2005) Modeling growth of *Clostridium perfringens* in
765 pea soup during cooling. Risk Anal 25:61–73
- 766 del Pulgar J, Roldán M, Ruiz J (2013) Volatile compounds profile of sous-vide cooked pork cheeks
767 as affected by cooking conditions (vacuum packaging, temperature and time). Molecules
768 18:12538–12547
- 769 Diaz P, Nieto G, Garrido MD, Sancho Banon S (2008) Microbial, physical–chemical and sensory
770 spoilage during the refrigerated storage of cooked pork loin processed by the sous vide method.
771 Meat Sci 80:287–292
- 772 Díaz P, Garrido MD, Bañón S (2010) The effects of packaging method (vacuum pouch vs. plastic
773 tray) on spoilage in a cook-chill pork-based dish kept under refrigeration. Meat Sci 84:538–544
- 774 Drummond L, Sun DW (2008) Immersion vacuum cooling of cooked beef safety and process
775 considerations regarding beef joint size. Meat Sci 80:738–743

- Edwards JS, Hartwell HJ (2006) Hospital food service: a comparative analysis of systems and introducing the 'Steamplicity' concept. *J Hum Nutr Diet* 19:421–430 777
- Elansari AM (2008) Hydrocooling rates of Barhee dates at the khalal stage. *Postharvest Biol Technol* 48:402–407 778
- Evans J, Russell S, James S (1996) Chilling of recipe dish meals to meet cook-chill guidelines. *Int J Refrig* 19:79–86 781
- Fennema OR, Powrie WD, Marth EH (1973) Low temperature preservation of foods and living matters. Marcel Dekker, New York, p 598 782
- French Ministry for Food, Agriculture and Fisheries (2003) Alimentation: Sécurité sanitaire: La restauration commerciale et collective à caractère social. <http://agriculture.gouv.fr/restauration-collective> 783
- Fuchigami M, Hyakumoto N, Miyazaki K (1995) Programmed freezing affects texture, pectin composition and electron microscopic structures of carrots. *J Food Sci* 60:137–141 784
- Garrett K (2012) Advantages and disadvantages of cook-chill food production within hospital setting. *NDFS* 445 785
- Gaze JE, Brown GD (1990) Determination of the heat resistance of a strain of non-proteolytic *Clostridium botulinum* type B and strain of type E, heated in cod and carrot homogenate over the temperature range 70 to 92°C. CFDR Technical Memorandum, No. 592, Campden Food and Drink Research Association, Chipping Campden 786
- Gaze JE, Brown GD, Gaskell DE, Banks JG (1989) Heat resistance of *Listeria monocytogenes* in homogenates of chicken, beef steak and carrot. *Food Microbiol* 6:251–259 787
- Golden NJ, Crouch EA, Latimer H, Kadry AR, Kaue J (2009) Risk assessment for *Clostridium perfringens* in ready-to-eat and partially cooked meat and poultry products. *J Food Prot* 72:1376–1384 788
- Gould GW (1999) Sous vide food: conclusions of an ECFF botulinum working party. *Food Control* 10:47–51 789
- Grant IR, Patterson MF (1995) Combined effect of gamma radiation and heating on the destruction of *Listeria monocytogenes* and *Salmonella typhimurium* in cook-chill roast beef and gravy. *Int J Food Microbiol* 27:117–128 790
- Greathouse KR, Gregoire MB, Nassar RF, Richards V, Marian C, Spears MC (1989) Comparison of conventional, cook-chill, and cook-freeze foodservice systems. *J Am Diet Assoc* 89:1606–1612 791
- Hiemstra SJ, Kim WG (1995) Factors affecting expenditures for food away from home in commercial establishments by type of eating place and meal occasion. *Hosp Res J* 19:15–31 792
- Hill MA (1994) Vitamin retention in microwave cooking and cook-chill foods. *Food Chem* 49:131–136 793
- Hu Z, Sun D-W (1999) The temperature distribution of cooked meat joints in an air-blast chiller during cooling process: CFD simulation and experimental verification. In: Paper presented at the 20th international congress of refrigeration, Sydney, Australia 794
- [AU3] Hu Z, Sun D-W (2000) Modelling of an experimental air-blast freezer using CFD code. In: Advance in refrigeration system, food technologies and cold chain, International Institute of Refrigeration, Paris, France, pp 395–400 795
- Ilicali C, Teik TH, Shian LP (1999) Improved formulations of shape factors for the freezing and thawing time prediction of foods. *Lebensm Wiss Technol* 32:312–315 796
- Jaloustre S (2011) [Quantitative risk assessment as a mean for evaluating hazard control measures in a food sector. Application to *Clostridium perfringens* in hospital catering] (in French). Ph.D. thesis: Paris Institute of Technology for Life, Food and Environmental Sciences (AgroParisTech) (France) 797
- James SJ (1990) Cooling systems for ready meals and cooked products. In: Field RW, Howell JA (eds) Process engineering in the food industry, 2. Convenience foods and quality assurance. Elsevier Applied Science, London, pp 88–97 798
- James SJ, James C (2005) Minimal processing of ready meals. In: Sun D-W (ed) Emerging technologies for food processing. Elsevier Academic, London, pp 717–732 799
- Jang JD, Seo GH, Lyu ES, Yam KL, Lee DS (2006) Hurdle effects of vinegar and sake on Korean seasoned beef preserved by sous vide packaging. *Food Control* 17:171–175 800

- 831 Jekanowski MD, Binkley JK, James Eales J (2001) Convenience, accessibility, and the demand for
832 fast food. *J Agric Resour Econ* 26:58–74
- 833 Kader AA (2002) *Postharvest technology of horticultural crops*. 3rd edn. University of California,
834 Agriculture and Natural Resources, Publication 3311, 535 p
- 835 Kalichevsky MT, Knorr D, Lillford PJ (1995) Potential food applications of high-pressure effects
836 on ice-water transitions. *Trends Food Sci Technol* 6:253–258
- 837 Kaushal P, Sharma HK (2012) Concept of computational fluid dynamics (CFD) and its applications
838 in food processing equipment design. *Food Process Technol* 3:1. doi:[10.4172/2157-
839 7110.1000138](https://doi.org/10.4172/2157-7110.1000138)
- 840 Kennedy C (2003) Developments in freezing. In: Zeuthen P, Bøgh-Sørensen L (eds) *Food preser-
841 vation techniques*. Woodhead Publishing, Cambridge, pp 228–240
- 842 Kim GT, Koo KM, Paik HD, Lyu ES, Lee DS (2001) Processing and storage of spinach products
843 using cook-chill and sous vide methods. *J Korean Soc Food Sci Nutr* 30:1095–1101
- 844 Koke K, Houska M, Kyhos K, Landfeld A (2002) Use of a computer program for parameter sensi-
845 tivity studies during thawing of foods. *J Food Eng* 52:219–225
- 846 Krebbers B, Matser AM, Koets M, Van den Berg RW (2002) Quality and storage-stability of high-
847 pressure preserved green beans. *J Food Eng* 54:27–33
- 848 Kuitche A, Daudin JD (1996) Modelling of temperature and weight loss kinetics during meat chill-
849 ing for time-variable conditions using an analytical-based method—I. The model and its sensi-
850 tivity to certain parameters. *J Food Eng* 28:55–84
- [AU4] 851 Lacey SL, Buckingham SE (1993) Isolation of *Salmonella enteritidis* from cook-chill food distrib-
852 uted to hospital patients. *J Hosp Infect* 25:133–136
- 853 Lee KE, Kim HJ, An DS, Lyu ES, Lee DS (2008) Effectiveness of modified atmosphere packaging
854 in preserving a prepared ready-to-eat food. *Packag Technol Sci* 21:417–423
- 855 Lee K-A, Kim K-T, Paik H-D (2011) Physicochemical, microbial, and sensory evaluation of cook-
856 chilled Korean traditional rice cake (Backseolgi) during storage via various packaging meth-
857 ods. *Food Sci Biotechnol* 20:1069–1074
- 858 Leistner L, Gould GW (2002) *Hurdle technology: combination treatments for food stability, safety
859 and quality*. Kluwer, New York, pp 65–89
- 860 Levkane V, Brasava DM, Dukalska L (2010) The shelf life of Sous vide packaged ready-to-eat
861 salad with meat in mayonnaise. *LLU Raksti* 25:78–88
- 862 Li B, Sun DW (2002) Effect of power ultrasound on freezing rate during immersion freezing of
863 potatoes. *J Food Eng* 55:277–282
- 864 Light N, Walker A (1990) *Cook-chill catering: technology and management*. Elsevier Applied
865 Science, London, pp 157–178
- 866 Manchester A, Clauson A (1994) Spending for food away from home outpaces food at home. In:
867 Food review, USDA/Economic Research Service, Washington DC, pp 12–15, July 1995 (and
868 subsequent updates)
- 869 Maroulis ZB, Kiranoudis CT, Marinos-Kouris D (1995) Heat and mass transfer modelling in air
870 drying of foods. *J Food Eng* 26:113–130
- 871 Martínez-Hernández GB, Artés-Hernández F, Colares-Souza F, Gómez PA, García-Gómez P, Artés
872 F (2013) Innovative cooking techniques for improving the overall quality of a Kailan-hybrid
873 broccoli. *Food Bioprocess Technol* 6:2135–2149
- 874 Martins RC, Almeida MG, Silva CLM (2004) The effect of home storage conditions and packag-
875 ing materials on the quality of frozen green beans. *Int J Refrig* 27:850–861
- 876 McDonald K, Sun D-W, Kenny T (2000) Comparison of the quality of cooked beef products
877 cooled by vacuum cooling and by conventional cooling. *LWT-Food Sci Technol* 33:21–29
- 878 McErlain L, Marson H, Ainsworth P, Burnett S (2001) Ascorbic acid loss in vegetables: adequacy
879 of a hospital cook-chill system. *Int J Food Sci Nutr* 52:205–211
- 880 Newton P, Gillespie, A (2010) A practical guide to vacuum packing. [www.lava-vacuum-packing.
881 com](http://www.lava-vacuum-packing.com), www.lava-europe.com
- 882 NFSMI (2002) *A guide to centralised foodservice systems*. National Food Services Management
883 Institute, University of Mississippi, Mississippi
- 884 Nyati H (2000) An evaluation of the effect of storage and processing temperatures on the microbio-
885 logical status of sous vide extended shelf life products. *Food Control* 11:471–476

- O'Leary E, Gormley TR, Butler F, Shilton N (2000) The effect of freeze-chilling on the quality of ready-meal components. *Lebensmittel-Wissenschaft und-Technologie* 33:217–224
- Olds DA, Mendonca AF, Sneed J, Bisha B (2006) Influence of four retail food service cooling methods on the behaviour of *Clostridium perfringens* ATCC 10388 in turkey roasts following heating to an internal temperature of 74°C. *J Food Prot* 69:112–117
- Park SH, Ryu HS, Hong GP, Min SG (2006) Physical properties of Frozen Pork Thawed by high pressure assisted thawing process. *Food Sci Technol Int* 12:347–352
- Peck MW, Goodburn KE, Betts RP, Stringer SC (2006) *Clostridium botulinum* in vacuum packed (VP) and modified atmosphere packed (MAP) chilled foods, final project report (b13006). Technical report. Institute of Food Research
- Poumeyrol G, Morelli E, Noel V, Cornu M (2012) Impact of the method chosen for measuring temperatures on the efficacy of rapid cooling of foods in catering facilities. *Food Control* 23:345–350
- Préstamo G, Fuster C, Risueño MC (1998) Effects of blanching and freezing on the structure of carrot cells and their implications for food processing. *J Sci Food Agric* 77:222–229
- Rahman AR, Henning WL, Westcott DE (1971) Histological and physical changes in carrots as affected by blanching, cooking, freezing, freeze-drying and compression. *J Food Sci* 36:500–502
- Rajkovic A, Smigic N, Devlieghere F (2010) Contemporary strategies in combating microbial contamination in food chain: review. *Int J Food Microbiol* 141:S29–S42
- Ramos M, Sanz PD, Aguirre-Puente J, Posado R (1994) Use of the equivalent volumetric enthalpy variation in non-linear phase-change processes: freezing zone progression and thawing-time determination. *Int J Refrig* 17:374–380
- Redmond GA, Gormley TR, Butler F, Dempsey A, Oxley E, Gerety A (2004) Freeze-chilling of ready-to-eat meal components. End-of-project report, Dublin, Ireland: Ashtown Food Research Centre (Teagasc), p 32, ISBN 1-84170-369-9
- Rodgers S (2003) Potential applications of protective cultures in cook-chill catering. *Food Control* 14:35–42
- Rodgers S (2004) Novel approaches in controlling safety of cook-chill meals. *Trends Food Sci Technol* 15:366–372
- Rozier J, Carlier V, Bolnot F, Tassin P, Rouve G, Gauthier M (1990) Plats cuisinés à l'avance et cuisson sous vide. *Maîtrise de la qualité hygiénique*. APRIA-CDIUPA, 72 p
- Rybka S (2000) Sous vide foods: the European perspective. *Food Aust* 53:451–454
- Rybka S, Kailasapathy K, Bergan J, Poniman S, Mikhail S, Gunasekara C, Lin Y, Ferraris J (2001) Storage characteristics of extended shelf life cook-chill meals. *Food Aust* 55:191–195
- Rybka-Rodgers S (1999) Developing a HACCP plan for extended shelf- life cook- chill ready- to-eat meal. *Food Aust* 51:430–433
- Rybka-Rodgers S (2001) Improvement of food safety design of cook-chill foods. *Food Res Int* 34:449–455
- Salvadori VO, Mascheroni RH (1991) Prediction of freezing and thawing times of foods by means of a simplified analytical method. *J Food Eng* 13:67–78
- Schafheitle JM, Light ND (1989) Sous-vide preparation and chilled storage of chicken ballotine. *Int J Food Sci Technol* 24:199–205
- Schellekens W, Martens T (1992) “Sous vide” cooking. ALMA K.U. Leuven, EUR 15018 EN EC Commission, Section DG XII, Brussels
- Schmidt CF, Lechowich RV, Folinazzo JF (1961) Growth and toxin production by type E *Clostridium botulinum*. *J Food Sci* 26:626–630
- Shakila RJ, Jeyasekaran G, Vijayakumar A, Sukumar D (2009) Microbiological quality of sous-vide cook chill fish cakes during chilled storage (3°C). *Int J Food Sci Technol* 44:2120–2126
- Shakila RJ, Raj BE, Felix N (2012) Quality and safety of fish curry processed by sous vide cook chilled and hot filled technology process during refrigerated storage. *Food Sci Technol Int* 18:261–269
- Shanaghy N, Murphy F, Kennedy K (1993) Improvements in the microbiological quality of food samples from hospital cook-chill system since the introduction of HACCP. *J Hosp Infect* 23:305–314

- 940 Sheard MA, Rodger C (1995) Optimum heat treatments for 'sous vide' cook-chill products. Food
941 Control 6:53–56
- 942 Simpson MV, Smith JP, Ramaswamy HS, Simpson BK, Ghazala S (1994) Thermal resistance of
943 *Streptococcus faecium* as influenced by pH and salt. Food Res Int 27:349–353
- 944 Stea TH, Johansson M, Jägerstad M, Frølich W (2006) Retention of folates in cooked, stored and
945 reheated peas, broccoli and potatoes for use in modern large-scale service systems. Food Chem
946 101:1095–1107
- 947 Stolle-Smits T, Beekhuizen JG, Recourt K, Voragen AGJ, Van Dijk C (1997) Changes in pectic and
948 hemicellulosic polymers of green beans (*Phaseolus Vulgaris L.*) during industrial processing. J
949 Agric Food Chem 45:4790–4799
- 950 Sun D-W, Hu Z (2002) CFD predicting the effects of various parameters on core temperature and
951 weight loss profiles of cooked meat during vacuum cooling. Comput Electron Agric 34:
952 111–127
- 953 Tansey F, Gormley R, Butler F (2010) The effect of freezing compared with chilling on selected
954 physico-chemical and sensory properties of sous vide cooked carrots. Innov Food Sci Emerg
955 Technol 11:137–145
- 956 Tragethon D (2011) Vacuum cooling—the science and practice. International Institute of Ammonia
957 Refrigeration annual meeting, Paper 36, Orlando, FL, USA, pp 1–22
- 958 USDA (2001) Performance standards for the production of meat and poultry products: proposed
959 rule. Federal Register, 66, 12589e12636, U.S. Department of Agriculture, Food Safety and
960 Inspection Service
- 961 Van Boekel MAJS (1999) Testing of kinetic models: usefulness of the multiresponse approach as
962 applied to chlorophyll degradation in foods. Food Res Int 32:261–269
- 963 Vomvoris W (2005) Cook-chill heats up. Foodserv Dir 18:58–64
- 964 Wang LJ, Sun D-W (2002) Modelling vacuum cooling process of cooked meat part 1: analysis of
965 vacuum cooling system. Int J Refrig 25:854–861
- 966 Werlein HD (1998) Comparison of the quality of sous-vide and conventionally processed carrots.
967 Zeitschrift-fur-Lebensmittel-Untersuchung-und-Forschung-A 207:311–315
- 968 Wilkinson PJ, Dart SP, Hadlington CJ (1991) Cook-chill, cook-freeze, cook-hold, sozss vide: risks
969 for hospital patients. J Hosp Infect 18:222–229
- 970 Xia B, Sun D-W (2002) Application of computational fluid dynamics (CFD) in the food industry:
971 a review. Comput Electron Agric 34:5–24
- 972 Xie G (2000) Comparison of textural changes of dry peas in sous vide cook-chill and traditional
973 cook-chill systems. J Food Eng 43:141–146
- 974 Yang MF (1990) An assessment of cook-chill foodservice systems. Master of Science in Nutrition
975 and Food Management, Oregon State University
- 976 Zhang Z, Sun D-W (2006) Effect of cooling methods on the cooling efficiencies and qualities of
977 cooked broccoli and carrot slices. J Food Eng 77:320–326
- 978 Zwietering MH, Stewart CM, Whiting RC (2010) Validation of control measures in food chain
979 using the FSO concept. Food Control 21:1716–1722

Author Queries

Chapter No.: 7 0002200104

Queries	Details Required	Author's Response
AU1	"Zheng and Sun 2004" is cited in text but not given in the reference list. Please provide details in the list.	
AU2	The citation "Jaloustre et al. 2011" (original) has been changed to "Jaloustre 2011". Please check if appropriate.	
AU3	Please provide complete details for the reference "Hu and Sun (2000)"	
AU4	Please provide in-text citation for the reference "Lacey and Buckingham (1993)".	

Uncorrected Proof

7.1 Giriş (Introduction)

İnsanlık tarihi boyunca, **gıda koruma yöntemleri**, kıtlık ve açlık dönemlerinde hayatta kalmayı sağlamada kritik bir rol oynamıştır. Ayrıca bu yöntemler, orduların, denizcilerin ve kaşiflerin uzun süre dayanıklı yiyecekler taşıyabilmesini sağlayarak **insanlık tarihini değiştiren gelişmelere** yol açmıştır.

Gıda koruma ve işleme alanındaki gelişmeler, insanlık tarihinde dört temel döneme ayrılabilir:

1. **Güvenli gıda üretimi dönemi** (ilk insan denemeleri – 1950’ler): Gıdaların bozulmadan, hastalık yapmadan saklanabilmesi öncelikti.
2. **Güvenli ve duyuşsal olarak hoş gıda üretimi dönemi** (1960–1980): Gıdanın sadece güvenli değil, aynı zamanda **görünüm ve tat açısından da çekici** olması hedeflendi.
3. **Besleyici ve kaliteli gıda üretimi dönemi** (1990–2000): Gıdaların hem güvenli hem de **besin değeri açısından dengeli** olması vurgulandı.
4. **Fonksiyonel ve yüksek kaliteli gıda üretimi dönemi** (2000–günümüz): Gıdaların belirli **sağlık ve fonksiyonel faydalar sağlaması** beklenmektedir.

Dolayısıyla gıda endüstrisi, **kurutma ve tuzlama** gibi ilkel yöntemlerden **konserveleme, soğutma ve dondurma** teknolojilerine doğru evrilmiştir. Amaç, sağlıklı ve besleyici gıdaya olan talebi karşılamaktır.

1960’larda, özellikle dışarıda yemek yeme alışkanlığının artmasıyla, **hazır ve hızlı servis edilebilen yemeklere** olan talep arttı. Bu durum, **pişir-soğut (cook-chill)** ve **pişir-dondur (cook-freeze)** sistemlerinin gelişimini teşvik etti. 1960–1970 arasında bu teknolojiler ticari ölçekte uygulanmaya başlandı. Ancak 1970–1980 döneminde elektrik, iş gücü ve ekipman maliyetlerinin artması nedeniyle bu sistemlere olan ilgi azaldı.

1990’lara gelindiğinde, dışarıda tüketilen yemek harcamaları ABD’de iki katına çıktı. Ayrıca kadınların iş gücüne katılımının artması, uzun çalışma saatleri ve zaman kısıtı gibi demografik değişimler, **hazır ve pratik yemeklere** olan ilgiyi artırdı. Bunun sonucunda, **enerji verimli pişirme-soğutma sistemlerinin** yaygınlaşmasıyla **cook-chill** ve **cook-freeze** teknolojileri yeniden gündeme geldi.

7.2 Cook-Chill ve Cook-Freeze Sistemleri

“Cook–Chill” ve “Cook–Freeze” sistemleri, gıdaların **geleneksel pişirme yöntemleriyle pişirilip**, ardından **mikrobiyolojik güvenlik ve kaliteyi koruyacak şekilde hızla soğutulması veya dondurulması** prensibine dayanır.

Amaç, **pişirme sonrası gıdanın duyuşsal özelliklerini ve besin değerini** en az kayıpla koruyarak, **servis zamanında yeniden ısıtılarak tüketime hazır hale getirilmesidir**.

◆ Cook–Chill (Pişir-Soğut) Sisteminin Temel Aşamaları

1. Hazırlık ve Pişirme:

- Yemekler tıpkı klasik yöntemle hazırlanır ve genellikle 70–90 °C’ye kadar pişirilir.
- Gıdanın iç sıcaklığı **en az 70 °C** olmalıdır (zararlı mikroorganizmaların yok edilmesi için).

2. Şok Soğutma (Chilling):

- Pişirme tamamlandıktan sonra yemek, **2 saat içinde +63 °C’den +3 °C’ye** düşürülmelidir.
- Bu işlem için **blast chiller (şok soğutucu)** kullanılır.
- Soğutma süresi **maksimum 90 dakika** olmalıdır.

3. Depolama:

- Soğutulmuş ürünler **+0 ile +3 °C arasında** muhafaza edilir.
- Bu sıcaklıkta gıdalar **5–7 gün boyunca** güvenli şekilde saklanabilir.

4. Dağıtım ve Yeniden Isıtma:

- Ürünler, servis noktasına soğuk zincir korunarak taşınır.
- Servis öncesi **yeniden ısıtma (reheating)** sırasında gıdanın çekirdek sıcaklığı **en az 70 °C’ye** çıkarılmalıdır.
- Ürün bir defadan fazla yeniden ısıtılmamalıdır.

◆ Cook–Freeze (Pişir-Dondur) Sisteminin Temel Aşamaları

1. Hazırlık ve Pişirme:

- Cook-chill ile aynıdır. Gıdalar tam pişirilir.
- Ancak hedef raf ömrü daha uzundur (3–6 ay).

2. Şok Dondurma (Blast Freezing):

- Pişirme sonrası 30 dakika içinde **–18 °C veya altına** düşürülmelidir.

- Bu işlem sırasında **büyük buz kristali oluşumunu engellemek için** hızlı dondurma tercih edilir.
- Bu, gıdanın dokusunun ve besin değerinin korunmasını sağlar.

3. Depolama:

- Ürünler **-18 °C ile -25 °C arasında** saklanır.
- Raf ömrü, gıdanın cinsine göre **3 ila 12 ay** arasında değişir.

4. Çözündürme ve Isıtma:

- Servis öncesi, dondurulmuş yemekler **+3 °C’de 24 saat bekletilerek çözündürülür.**
- Ardından **70 °C’ye kadar** ısıtılır.
- Dondurulmuş ürünler **mikrodalga veya konveksiyon fırınlarında** çözündürülüp ısıtılabilir.

Sistemlerin Karşılaştırması

Özellik	Cook-Chill	Cook-Freeze
Saklama Sıcaklığı	+0 °C – +3 °C	-18 °C – -25 °C
Raf Ömrü	5–7 gün	3–12 ay
Ekipman Gereksinimi	Şok soğutucu	Şok dondurucu
Enerji Tüketimi	Daha az	Daha fazla
Doku ve Tat Kalitesi	Çok iyi	İyi
Uygun Olduğu Ürünler	Çorba, sos, sebze yemekleri	Etli ve sulu yemekler, fırın ürünleri
Maliyet	Düşük-Orta	Orta-Yüksek

Temel Prensip

Cook-chill sisteminde **amaç, ürünü kısa sürede servis edebilmek**; cook-freeze sisteminde ise **raf ömrünü maksimuma çıkarmaktır**.

Her iki sistemde de en kritik nokta, “soğutma/dondurma hızının yeterli olması” ve soğuk zincirin kesintisiz sürdürülmesidir.

7.3 Mikrobiyolojik ve Gıda Güvenliđi Hususları

Cook-chill ve cook-freeze sistemlerinin başarısı, **mikroorganizmaların kontrol altına alınmasına** bađlıdır.

Amaç, pişmiş gıdada mikroorganizma sayısını minimuma indirip, sođutma ve depolama sırasında **yeniden çođalmalarını engellemektir**.

♦ 1. Pişirme Aşaması – “Öldürme (Kill) Noktası”

- Pişirme, gıdadaki patojen mikroorganizmaların yok edilmesini sağlar.
- Gıdanın **çekirdek sıcaklığı 70 °C’nin üzerine** çıkmalı, bu sıcaklık **en az 2 dakika** sürmelidir.
- Özellikle *Salmonella spp.*, *Listeria monocytogenes*, *Clostridium perfringens* ve *Bacillus cereus* gibi bakteriler bu aşamada etkisiz hale gelir.
- Ancak **ısıya dayanıklı sporlar** (özellikle *C. perfringens* sporları) tamamen ölmez. Bu nedenle, pişirme sonrası **sođutma süresi ve hızı** hayati önem taşır.

♦ 2. Sođutma Aşaması – “Kritik Kontrol Noktası”

- Cook-chill sisteminde en kritik evredir.
- Pişirme sonrası ürünün 60 °C’den 10 °C’ye inmesi sırasında **bakteriyel sporlar aktifleşebilir**.
- Bu yüzden bu sıcaklık aralığında geçen süre **2 saati geçmemelidir** (ideal: 90 dakika).
- Sođutma, **blast chiller (şok sođutucu)** ile yapılmalı; bu cihaz, gıdanın merkez sıcaklığını kısa sürede düşürerek bakteriyel büyümeyi durdurur.

✿ *Ev tipi buzdolabı veya sođuk hava odaları bu işlem için uygun değildir çünkü yeterince hızlı sođutma sağlayamazlar.*

♦ 3. Depolama – “Büyümeyi Durdurma Noktası”

- Cook-chill ürünleri **0–3 °C’de**, cook-freeze ürünleri **–18 °C veya altında** saklanmalıdır.
- Sođutulmuş ürünlerde *Listeria monocytogenes* düşük sıcaklıkta bile yavaş da olsa çođalabilir. Bu nedenle depolama süresi **maksimum 7 gün** olmalıdır.

- Dondurulmuş ürünlerde bakteriler canlı kalabilir ama **çoğalamazlar**; çözünme sonrası tekrar aktifleşirler.

✚ Bu yüzden cook-freeze ürünlerde **çözülme sonrası 24 saat içinde ısıtılıp servis edilmesi** gerekir.

♦ 4. Yeniden Isıtma – “Güvenli Servis Noktası”

- Gıda, servis öncesinde **en az 70 °C’ye kadar ısıtılmalıdır**.
- *Clostridium perfringens* sporlarının öldürülmesi için bu sıcaklık gereklidir.
- Yeniden ısıtılan yemek **tek seferde tüketilmeli**, ikinci kez ısıtılmamalıdır.
- Yeniden ısıtma işlemi mikrodalga, konveksiyon fırın veya buharlı fırınla yapılabilir; ancak ısı dağılımının **tüm porsiyonda homojen** olması sağlanmalıdır.

♦ 5. Hijyen ve Ekipman Temizliği

- Cook-chill/cook-freeze üretimi, **yüksek hijyen standardı** gerektirir.
- Çalışma alanları, personel ve ekipman, **HACCP prensiplerine** uygun olmalıdır.
- Şok soğutucu, dondurucu, depolama ve paketlenme alanları arasında **çapraz bulaşmayı (cross-contamination)** önlemek için net bir **iş akışı yönü** (pişmemişten pişmişe doğru) oluşturulmalıdır.

⚠ Kritik Sıcaklık Aralıkları

Aşama	Sıcaklık Aralığı	Maksimum Süre	Risk
Pişirme	$\geq 70\text{ °C}$	2–5 dk	Patojenlerin ölmesi
Soğutma	$60\text{ °C} \rightarrow 10\text{ °C}$	≤ 2 saat	Spor aktivasyonu riski
Depolama (Cook-Chill)	$0-3\text{ °C}$	≤ 7 gün	Listeria riski
Dondurma	-18 °C veya altı	3–12 ay	Sporlar uyur halde
Yeniden Isıtma	$\geq 70\text{ °C}$	–	Güvenli servis sıcaklığı

♦ Sonuç

Cook-chill ve cook-freeze sistemlerinin güvenliği;

- 1) yeterli pişirme,
- 2) hızlı soğutma,
- 3) soğuk zincirin korunması ve
- 4) tek seferlik ısıtma aşamalarının eksiksiz yapılmasına bağlıdır.

Bu işlemler doğru yürütülmezse, gıdalar **mikrobiyolojik olarak tehlikeli hale gelebilir**.

Özellikle *Listeria* ve *Clostridium* türleri bu sistemlerde en sık rastlanan risk faktörleridir.

7.4 Kalite, Besin Değeri ve Duyusal Özelliklerdeki Değişimler

Cook-chill ve cook-freeze sistemlerinin başarısı yalnızca **mikrobiyolojik güvenliğe** değil, aynı zamanda gıdanın **duyusal kalitesinin korunmasına** da bağlıdır.

Bu sistemlerdeki temel amaç, **pişirme ile servis arasındaki süreyi uzatırken** ürünün tadı, kokusu, görünümü ve besin içeriğini mümkün olduğunca korumaktır.

♦ 1. Tat ve Aroma Değişimleri

- Pişirme sonrası hızlı soğutma, gıdadaki uçucu aroma bileşenlerinin korunmasına yardımcı olur.
- Ancak yetersiz soğutma veya uzun süreli bekleme, **oksidasyon ve Maillard reaksiyonlarını** artırarak **yağ acılaşması** ve **renk kararmasına** neden olur.
- Özellikle etli yemeklerde (ör. orman kebabı, köfte) **proteinlerin yeniden ısıtma sırasında denatürasyonu**, tat yoğunluğunda azalmaya yol açabilir.
- **Cook-freeze** sisteminde, dondurma ve çözündürme işlemleri aromatik bileşenlerde %10–20 kayba neden olabilir.

📌 Öneri:

Yağlı et yemeklerinde tereyağ yerine sıvı yağ kullanmak, dondurulmuş ürünlerde aroma kaybını azaltır.

♦ 2. Renk ve Görünüm

- Sebzelerde ve soslu yemeklerde en belirgin değişim, **klorofilin bozunmasıyla yeşilin kahverengileşmesidir**.
- Dondurulmuş ürünlerde, özellikle yeniden ısıtıldığında **pigment oksidasyonu** meydana gelir.

- Cook-chill sisteminde 3–4 gün boyunca renk deęiřimi minimaldir.
- Ancak cook-freeze sisteminde çözölme sonrası renk farkı belirgindir.

Öneri:

Renk kaybını azaltmak için řok soęutma öncesi kısa süreli **blanching (hařlama)** veya **pH dengeleme (limon-tuz karıřımı)** uygulanabilir.

♦ 3. Doku (Kıvam) Deęiřimleri

- Piřirme sonrası hızlı soęutulan ürünlerde hücre yapısı korunur, ancak yavaş soęutulanlarda **su kaybı ve doku yumuřaması** olur.
- Dondurulmuř gıdalarda büyük buz kristalleri hücre zarlarını yıkar, bu da çözöndüğünde **sulanmıř, daęılmıř yapı** oluřturur.
- Cook-chill ürünleri daha “taze kıvamlı”, cook-freeze ürünleri ise “biraz sulu” olur.

Öneri:

- Çorba, sos ve pilavlarda niřasta oranını %5–10 artırmak (ör. 1 litreye 10 g un/niřasta eklemek) doku stabilitesini korur.
 - Etli yemeklerde piřirme sonrası hemen porsiyonlama yapılması, kas liflerinin daęılmasını önler.
-

♦ 4. Besin Deęeri Deęiřimleri

- Piřirme sırasında **vitamin C, folik asit ve bazı B vitaminleri** ısıya duyarlıdır.
- Soęutma/dondurma sürecinde besin kaybı genelde **%5–15 arasında** olur.
- Yaęda çözönen vitaminler (A, D, E, K) nispeten stabildir.
- Protein ve mineral kaybı ise minimum düzeydedir.

Besin Öęesi	Piřirme Kaybı	Cook-Chill Kaybı	Cook-Freeze Kaybı
C Vitamini	%30–40	%45	%55
Folik Asit	%20–30	%35	%50

Protein	%5	%7	%10
Demir	%2	%3	%5
Kalsiyum	%2	%3	%4

Öneri:

- Çorba ve sebze yemeklerinde **pişirme suyunun atılmaması**, besin kaybını en aza indirir.
- Şok soğutma süresi ne kadar kısa olursa vitamin kaybı da o kadar az olur.

♦ 5. Duyusal Kalitenin Korunması İçin En İyi Uygulamalar

1. **Soğutmayı 90 dakika içinde tamamla.**
Yavaş soğutma en büyük kalite düşmanı.
2. **Porsiyonları pişirme sonrası hemen ayır.**
Toplu kaplarda bekleyen yemeklerde doku ve tat bozulur.
3. **Şeffaf ama oksijen bariyerli kaplar (PP, PET/CPET) kullan.**
Bu kaplar renk ve aroma kaybını %30 azaltır.
4. **Tekrar ısıtmayı hızlı yap.**
Mikrodalga veya buharlı fırın tercih et, ısıtma süresini 5 dakikayı geçirme.
5. **Dondurulmuş ürünleri çözülür çözülmez servis et.**
Çözülme sonrası bekletmek hem doku hem mikrobiyoloji açısından risklidir.

Sonuç

Cook-chill sistemi:

- **Taze tada en yakın sonuçları** verir.
- 5–6 günlük servis süresi için idealdir.

Cook-freeze sistemi:

- **Uzun raf ömrü sağlar** ancak aroma ve doku kaybı daha fazladır.

- Merkezi üretim – çoklu şube sistemi (örneğin franchise yapısı) için uygundur.

Her iki sistemde de en önemli kalite kriterleri:

şoklama süresi, ambalaj türü ve depolama sıcaklığının kararlılığıdır.

7.5 Ambalajlama, Ekipman ve Uygulama Örnekleri

Cook-chill ve cook-freeze sistemlerinde kullanılan ambalaj, sadece taşıma aracı değil, aynı zamanda **gıdanın raf ömrünü, güvenliğini ve görünümünü koruyan aktif bir bileşendir.**

Bu nedenle ambalaj malzemesi seçimi, **ısıya dayanıklılık, oksijen geçirgenliği, sızdırmazlık ve yeniden ısıtılabilirlik** gibi kriterlere göre yapılmalıdır.

♦ 1. Ambalaj Malzemeleri

Ambalaj Türü	Kullanım Alanı	Avantajlar	Dezavantajlar
PP (Polipropilen)	En yaygın tip (çorba, pilav, etli yemekler)	Uygun fiyat, mikrodalgaya dayanıklı, ısıya dirençli	Dondurma sonrası çatlama riski
CPET (Crystalline PET)	Cook-freeze sisteminde	-40 °C ile +220 °C arasında dayanıklı, fırın ve mikrodalgada kullanılabilir	Diğerlerine göre pahalı
Alüminyum Kaplar	Etli, fırınlı yemekler	Isıyı eşit dağıtır, ısıtması kolay	Mikrodalgada kullanılamaz, asidik gıdalarla reaksiyon riski
Vakum Torbaları (PA/PE veya Multilayer)	Soslar, çorbalar	Oksijen geçirmez, uzun raf ömrü sağlar	Doldurma ve mühürleme ekipmanı gerekir
Termoform Kaplar	Endüstriyel üretim hatlarında	Otomatik doluma uygun, estetik sunum	Ekipman yatırımı gerekir

📌 **Senin senaryonda (küçük işletme, porsiyon bazlı servis):**

👉 En uygun kombinasyon: **PP kap + şeffaf kapak + etiketli sleeve**

Bu kombinasyon hem şok soğutmaya hem mikrodalga ısıtmaya uygundur, hem de düşük maliyetlidir.

♦ 2. Ambalajlama Aşamaları

1. Pişirme sonrası ilk 15 dakika içinde porsiyonlama yapılmalıdır.

Gıdanın sıcaklığı hâlâ yüksekken ambalaja alınırsa, bakteriyel bulaşma riski azalır.

2. **Ambalaj doldurulurken hava boşluğu minimum olmalıdır.**

Oksijenli ortam, *Listeria* ve *Pseudomonas* gibi bakterilerin üremesini kolaylaştırır.

3. **Vakumlama veya modifiye atmosfer (MAP) sistemi kullanılabilir.**

MAP sisteminde genellikle şu gaz karışımları tercih edilir:

- %70 N₂ (azot) + %30 CO₂
- Bu kombinasyon, aerobik bakterilerin büyümesini durdurur ve raf ömrünü %30 artırır.

4. **Ambalajların üzerine üretim tarihi, son tüketim tarihi ve ısıtma talimatı basılmalıdır.**

Örneğin:

“+3 °C’de saklayınız. Mikrodalgada 3–4 dk ısıtınız. Yeniden dondurmayınız.”

♦ 3. Ekipman Gereksinimleri

Ekipman	İşlev	Açıklama
Blast Chiller (Şok Soğutucu)	Hızlı soğutma	90 dakikada +70 °C → +3 °C; HACCP için zorunlu
Blast Freezer (Şok Dondurucu)	Hızlı dondurma	–40 °C hava akımı; hücre dokusunu korur
Soğuk Hava Dolabı (0–3 °C)	Cook-chill depolama	7 güne kadar güvenli saklama
Derin Dondurucu (–18 °C)	Cook-freeze depolama	3–12 ay arası raf ömrü
Vakum Makinesi (opsiyonel)	Hava tahliyesi	Raf ömrünü uzatır
Etiket Yazıcı (inkjet veya termal)	Takip sistemi	Üretim tarihi ve parti kodu basımı
Mikrodalga / Konveksiyon Fırın	Servis öncesi ısıtma	70 °C hedef sıcaklık, 5 dk içinde ısıtma

✿ Küçük hacimli üretimlerde blast chiller ve dondurucu, **aynı gövdede iki kademeli cihaz** olarak da satın alınabilir.

Bu tür cihazlar 10.000–20.000 TL aralığında mini modeller olarak mevcuttur (ör. Polar, Sagi, Saro).

◆ 4. Ambalajlama Süreci Akış Şeması

Piştirme → Porsiyonlama → Ambalajlama → Şok Soğutma / Dondurma → Etiketleme → Depolama → Sevkiyat / Isıtma / Servis

Her aşama arasında **ürün sıcaklığına göre renk kodlu kasalar veya arabalar** kullanılması önerilir:

- Kırmızı: sıcak hat
- Mavi: soğutma hattı
- Beyaz: soğuk depolama hattı

Bu, çapraz bulaşmayı azaltır ve HACCP belgelerinde “kontrol izlenebilirliği” sağlar.

◆ 5. Gerçek Uygulama Örneği: İngiltere NHS Hastaneleri Cook–Chill Sistemi

İngiltere’deki büyük sağlık kuruluşları (ör. NHS Trust) 1990’lardan beri merkezi cook-chill sistemini kullanır.

Süreç şöyle işler:

1. Merkez mutfakta gıdalar pişirilir, +3 °C’ye kadar şok soğutulur.
2. Porsiyonlanıp PP kaplara doldurulur.
3. Her ürün soğuk zincirle (2–4 °C) taşınır.
4. Hastanelerde mikrodalga veya “regen oven” (yeniden ısıtma fırını) ile 15 dakikada servise hazır hale getirilir.

Bu sistemle:

- Ürün kaybı %10’un altına düşer,
 - Personel maliyeti %25 azalır,
 - Gıda güvenliği şeffaf biçimde izlenebilir hale gelir.
-

◆ 6. Senin İşletmen İçin Uygun Uygulama Modeli (Basitleştirilmiş Cook–Chill)

Aşama	Eylem	Önerilen Ekipman
1	Rasyo Catering yemekleri sıcak teslim alınır (+70 °C)	Isı yalıtımlı termos
2	Porsiyonlama yapılır (PP kaplara)	Çelik kepçe, PP kap
3	Şok soğutma (90 dk)	Mini blast chiller
4	Etiketleme + tarih	Termal yazıcı
5	Depolama +3 °C	Ticari buzdolabı
6	Servis günü mikrodalgada ısıtma	800W mikrodalga, 3 dk

📌 Bu süreçle, senin ürünlerin **5–6 gün boyunca taze ve güvenli** kalabilir. Eğer günlük tüketim hızı düşükse, aynı sistem “cook-freeze”e dönüştürülerek 2–3 ay raf ömrü sağlanabilir.

7.6 Cook-Chill ve Cook-Freeze Sistemlerinin Avantajları, Sınırlamaları ve Ekonomik Değerlendirmesi

Bu sistemlerin tercih edilmesinin ana nedeni, **merkezi üretimle gıda kalitesini korurken iş gücü, zaman ve maliyetten tasarruf** sağlamasıdır.

Ancak doğru uygulanmadığında yüksek yatırım maliyetleri, ekipman eksikliği ve kalite kayıpları söz konusu olabilir.

♦ 1. Avantajlar

✅ a. İş Gücü ve Zaman Verimliliği

- Gıdalar toplu şekilde üretildiği için **üretim planlaması kolaylaşır**.
- Günlük pişirme gereksinimi azalır; personel yükü %30-50 oranında düşer.
- Servis noktalarında yalnızca **ısıtma ve dağıtım personeli** gerekir.
- Tatil günleri veya ani sipariş artışlarında **hazır stoktan çalışma** mümkündür.

📌 Örnek:

Senin sisteminde 1000 porsiyonluk üretim tek seferde yapılabilir.

Soğutulmuş şekilde 5-6 gün saklanacağı için her gün ısıt-servis modeliyle günlük iş yükün 1/5’e iner.

✅ b. Kalite ve Standartlaşma

- Aynı reçete tekrarlandığı için lezzet ve görünüm sabit kalır.
- HACCP ve ISO 22000 gıda güvenliği sistemlerine entegre edilebilir.
- Pişirme koşulları, porsiyon gramajı ve sunum tamamen **kontrol altındadır**.

📌 *Senin menündeki yemekler (örn. İzmir köfte, mercimek çorbası, nohutlu pilav) bu sisteme çok uygun çünkü homojen, soslu ve yapısal bozulmaya dayanıklıdır.*

✅ c. Maliyet Avantajı

- Büyük hacimli üretimle **hammadde birim maliyeti düşer** (%10-15).
- Enerji tüketimi tek seferde toplu pişirmeyle azalır.
- Atık oranı %2-3 seviyesine iner.
- Personel ve gaz-elektrik tasarrufu sayesinde toplam işletme gideri %20-30 azalır.

📌 Örnek:

Günde 500 porsiyon yemek pişiren klasik bir mutfakın aylık enerji + personel gideri 60.000 TL iken, cook-chill sisteminde bu rakam 40.000 TL civarına düşebilir.

✅ d. Lojistik Esnekliği

- Gıdalar **soğuk zincirle 24-48 saat içinde farklı lokasyonlara** taşınabilir.
- Aynı üretim merkeziyle çok sayıda şube, büfe veya ofis beslenebilir.
- PP veya CPET kaplar sayesinde **ambalajdan direkt servis** yapılabilir.

📌 *Bu, senin “eczane, psikolog ofisi, kuaför gibi işletmelere sıcak menü satma” modelini doğrudan destekler.*

Yemekleri sabah şoktan çıkarıp mikrodalgada ısıtıp servis edebilirsin.

✅ e. Gıda Güvenliği

- Sıcak zincir yerine **soğuk zincir kullanımı**, bakteri üremesini neredeyse sıfırlar.
- Doğru sıcaklık izleme (termometre ve etiketleme) ile HACCP kayıtları kolay tutulur.

- Manuel müdahale azaldığı için **çapraz bulaşma riski** minimumdur.
-

♦ 2. Sınırlamalar ve Dikkat Edilmesi Gerekenler

⚠ a. Yüksek Başlangıç Yatırımı

- Şok soğutucu, dondurucu, paketlenme ve etiketleme sistemleri ilk yatırım maliyeti oluşturur.
- Küçük işletmeler için başlangıç maliyeti 100-250 bin TL civarındadır.
- Ancak 12-18 ay içinde bu yatırım kendini amorti eder.

⚠ b. Ürün Seçim Kısıtlaması

- Kızartmalar, çıtır hamurlu veya yumurtalı ürünler bu sistemlerde kıvam kaybeder.
- Sulu ve soslu yemekler (güveç, çorba, pilav) cook-chill için idealdir.
- Tatlılar ve mayalı ürünler (su böreği, poğaça vb.) dondurmaya uygun değildir.

📌 *Senin menüdeki tüm yemekler — etli, tavuklu, pilavlı ve çorbalı çeşitler — cook-chill sistemine uygundur.*

⚠ c. Ekipman Bağımlılığı

- Soğutucu arızası veya elektrik kesintisi ürün güvenliğini doğrudan etkiler.
- Her cihazda **sıcaklık kaydı ve alarm sistemi** olmalıdır.
- Aksi durumda tek bir hata 1000 porsiyonun ziyan olmasına yol açabilir.

⚠ d. Çalışan Eğitimi Gereksinimi

- Pişirme, şoklama, depolama ve ısıtma süreçleri **hijyen ve zaman açısından disiplin** ister.
 - Personelin HACCP ve temel gıda hijyeni eğitimi alması gerekir.
-

♦ 3. Ekonomik Değerlendirme (Özet)

Kalem	Klasik Günlük Üretim	Cook-Chill Sistemi	Fark
Üretim sıklığı	Her gün	2-3 günde 1	Zaman avantajı
Enerji tüketimi	%100	%70	–%30 tasarruf
Personel	5 kişi	3 kişi	–%40 tasarruf
Gıda fire oranı	%10	%3	–%7 tasarruf
Ortalama porsiyon maliyeti	45 TL	37 TL	–%18 düşüş
İlk yatırım	Düşük	Orta (200 bin TL civarı)	12-18 ay amortisman

♦ 4. Stratejik Uygunluk ve Genişleme Potansiyeli

Cook-chill sistemi;

- **Tek merkezli üretim,**
- **Birden fazla satış noktası (ofis, kafe, büfe) ve**
- **Küçük hacimli ama sabit menüli işletmeler için en uygundur.**

Cook-freeze sistemi ise;

- **Bölgesel dağıtım veya e-ticaret** modelinde (örn. dondurulmuş ev yemekleri satışı) avantaj sağlar.
- Ürünler –18 °C’de taşınabildiği için lojistik daha kolay ama enerji maliyeti daha yüksektir.

📌 *Senin modelinde (Üsküdar bölgesinde ofislere sıcak menü satışı), cook-chill sistemi en ideal seçenektir.*

Ancak ileri aşamada “donuk yemek satış” işine genişlemek istersen, sistem altyapısı buna uyumlu kalır.

♦ 5. Sonuç

Cook-chill ve cook-freeze sistemleri, modern catering ve küçük ölçekli ev yemekleri işletmeleri için **verimlilik, kalite ve gıda güvenliği** açısından büyük avantajlar sağlar.

Başarı için kilit faktörler:

- 1 Gıdaların uygun reçetelerle hazırlanması,
- 2 Şok soğutma/dondurma süresinin kontrolü,
- 3 Doğru ambalajlama ve etiketleme,
- 4 Soğuk zincir yönetimi,
- 5 Servis öncesi tek seferlik ısıtma protokolü.

Bu prensipler uygulandığında, hem **müşteri memnuniyeti yüksek**, hem de **operasyonel maliyet düşük** bir sistem kurulabilir.

♦ Ek A – Cook-Chill Sisteminde HACCP Planı Örneği

Bu plan, toplu üretim mutfaklarında uygulanan **tehlike analizleri ve kritik kontrol noktaları (HACCP)** sistemine göre hazırlanmıştır.

Senin işletmende de 3 temel tehlike vardır: **mikrobiyolojik, kimyasal ve fiziksel**.

Aşağıda 6 aşamalı HACCP örneği yer alıyor 📌

Aşama	Kritik Kontrol Noktası (CCP)	Kritik Limit	İzleme Yöntemi	Düzeltilici Faaliyet
1 Pişirme	Gıdanın iç sıcaklığı en az 75 °C olmalı	75 °C	Dijital termometre	Pişirme süresini uzat
2 Soğutma	90 °C → +3 °C aralığı 90 dk içinde geçilmeli	90 dk	Şok soğutucu sıcaklık kaydı	Soğutmayı hızlandır, parti ayır
3 Depolama	+3 °C veya altında saklama	+3 °C	Günlük buzdolabı termometresi	Ürünü imha et veya yeniden ısıt
4 Dondurma (Cook-Freeze)	-18 °C veya altında olmalı	-18 °C	Derin dondurucu göstergesi	Yeniden dondurma yapılamaz
5 Dağıtım	0-4 °C arasında nakliye	+4 °C	Taşıma termometresi	Ürünü iade et veya ısıtıp hemen servis et
6 Yeniden Isıtma	Servis öncesi merkez sıcaklık ≥ 70 °C	70 °C	Termometre / süre kontrolü	Tekrar ısıt, 2. kez ısıtma yapılmaz

📌 Not:

Bu tablo, Üsküdar'daki belediye denetimlerinde istenen “kritik kontrol planı” formatıyla da birebir uyumludur.

(Tarif sayfalarına ekleyebilirsiniz.)

♦ Ek B – Önerilen Şok Soğutma Süreleri (Ürün Türüne Göre)

Ürün	Miktar	Başlangıç Sıcaklığı	Hedef Sıcaklık	Tahmini Süre
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Mercimek / Ezogelin Çorbası	20 L	+85 °C	+3 °C	90 dk
Etlı Yemek (Orman Kebabı, Köfte vb.)	15 kg	+80 °C	+3 °C	75–90 dk
Pilav / Makarna	10 kg	+85 °C	+3 °C	60 dk
Sebze Yemekleri	10 kg	+80 °C	+3 °C	70 dk
Soslar (Domates, Beşamel vb.)	5 L	+90 °C	+3 °C	45 dk

⚠ Uyarı:

Ev tipi buzdolabı bu süreleri sağlayamaz.

Profesyonel **blast chiller** (şok soğutucu) veya **soğuk hava fanlı kabin** kullanmak gerekir.

Eğer elinde profesyonel cihaz yoksa:

- Çorbayı 2-3 L'lik kaplara paylaşır,
- Kabin dışını buzlu su dolu tepsiye koy,
- Ara sıra karıştır — soğutma süresi 2 saate kadar uzayabilir ama yine de güvenli olur.

♦ Ek C – Mini Ekipman Listesi (Küçük Ölçekli İşletmeler İçin)

Ekipman	Görev	Ortalama Fiyat (2025 itibarıyla)
Mini Blast Chiller (10 tepsi kapasiteli)	90 dk'da 20 kg soğutur	18.000 TL
Mini Blast Freezer (–40 °C)	30 kg dondurma kapasitesi	22.000 TL
Endüstriyel Buzdolabı (+2/+4 °C)	Cook-chill depolama	15.000 TL
PP Kap (1000 ml) + Şeffaf Kapak	Porsiyonlama	1,2 TL/adet
Termal Etiket Yazıcı	Etiketleme (üretim/son tüketim)	3.000 TL
Dijital Termometre (prob uçlu)	Sıcaklık ölçümü	500 TL
800 W Mikrodalga	Yeniden ısıtma	4.000 TL
Buzlu Soğutma Kabı (manuel soğutma için)	Geçici çözüm	1.500 TL

Toplam tahmini yatırım (cihaz + sarf malzeme) ≈ **60–65 bin TL**



Bu, senin gibi butik bir mutfağın **ilk yıl içinde kendi üretimini devralmasına** da olanak sağlar.

♦ Ek D – HACCP Takip Formu Örneği

Cook-Chill Üretim Günlük Takip Formu

Tarih	Ürün Adı	Piştirme Sıcaklığı (°C)	Soğutma Başlangıç	Soğutma Sonu	Süre (dk)	Depo Sıcaklığı	Kontrol Eden
01.11.2025	Mercimek Çorbası	85 °C	14:00	15:25	85 dk	+2 °C	Z.Keleş
01.11.2025	İzmir Köfte	82 °C	16:00	17:20	80 dk	+3 °C	Z.Keleş
02.11.2025	Tavuk Sote	78 °C	13:30	14:45	75 dk	+3 °C	Z.Keleş

📌 Bu formu bir Excel tablosu hâlinde tutarsan, belediye ruhsat denetiminde “soğuk zincir kayıtları mevcut” ibaresiyle işletmen puan alır.

♦ Ek E – Örnek Servis Prosedürü (Cook-Chill Ürünleri İçin)

- 1) Ürünleri buzdolabından çıkar.
- 2) Mikrodalga veya konveksiyon fırında merkez sıcaklık **70 °C** olana kadar ısıt.
- 3) Kapaklar delikli ise buhar çıkışı için açma yeri bırak.
- 4) Yeniden dondurma veya ikinci ısıtma kesinlikle yapılmaz.
- 5) Servis edilen gıda 2 saat içinde tüketilmelidir.

📌 **Pratik İpucu:**

Eğer dağıtımı sen yapıyorsan, taşıma termosu içine **ısı akülü plaka** (60 °C tutucu jel) koyarak servis süresini 2 saat uzatabilirsin.

♦ Ek F – Basitleştirilmiş Cook-Freeze Süreci (Donuk Ürünler İçin)

Aşama	Açıklama
Piştirme	Ürün iç sıcaklığı ≥ 75 °C
Ön Soğutma	30 dk oda sıcaklığında
Şok Dondurma	-40 °C’de 90 dk
Depolama	-18 °C’de 3–6 ay

Çözündürme	+3 °C’de 12 saat
Isıtma	70 °C merkez sıcaklıkta servis

📌 Bu sistemde **dondurma sonrası porsiyonlama yapılmaz** — çünkü çözülmüş ürün tekrar dondurulamaz.

O yüzden senin dediğin gibi, *pişmiş yemeği porsiyonlayıp sonra dondurmak* en mantıklı yoldur ✅

♦ Ek G – Cook-Chill ve Cook-Freeze Sistemlerinin Yasal Dayanakları (Türkiye)

Yönetmelik / Kaynak	Uygulama Alanı
Türk Gıda Kodeksi – Mikrobiyolojik Kriterler Yönetmeliği (2011)	Pişmiş ve soğutulmuş gıdalar için Listeria limitleri
Gıda İşletmelerinin Kayıt ve Onay Yönetmeliği (2020)	Üretim yeri ruhsatı ve HACCP zorunluluğu
ISO 22000:2018 Gıda Güvenliği Yönetim Sistemi	Tehlike analizi ve izlenebilirlik
Belediye Ruhsat Yönetmeliği (Üsküdar Belediyesi, 2025)	Pişmiş ürün satışı için “Hazır Yemek Satışı” ruhsatı şartı
TSE 13075 – Hazır Yemek Üretimi Standardı	Cook-chill ve cook-freeze dahil toplu yemek üretimi kuralları

🧠 Sonuç: Senin İşletmene En Uygun Model

Senin sistemin için en uygun model:

Cook-Chill + PP Porsiyonlama + Günlük Isıt-Servis

çünkü:

- Ürünlerin hepsi (çorba, etli, tavuklu, pilavlı yemekler) bu yönteme %100 uygun.
- Gıda güvenliği ve belediye denetimlerinde avantaj sağlar.
- Düşük yatırım, hızlı dönüş ve yüksek müşteri memnuniyeti sağlar.